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MILITARY SPECIFICATION

RADIO SET AN/GRC-106( ) (UNITS OF)

This specification is approved for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 This specification covers the following units of a high-frequency, single-sideband (SSB) radio receiving-transmitting set operating in the frequency range 2.0000 to 29.9999 megahertz (MHz) (see 6.1 and 6.8):

Receiver-Transmitter, Radio RT-834( )/GRC

Amplifier, Radio Frequency AM-3349( )/GRC-106

1.2 APPENDIX - The following appendix forms a part of this specification:

Appendix A - Reliability Test Plan for Radio Set AN/GRC-106( ).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

NN-P-71	Pallet, Materials-handling, Wood (General Construction Requirements)
QQ-S-781	Strapping, Flat; Steel
PPP-T-45	Tape, Gummed, Paper, reinforced and plain, for sealing and securing.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing)
PPP-T-97	Tape, Pressure-Sensitive, Filament Reinforced
PPP-P-291	Paperboard, Wrapping, Cushioning

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SPECIFICATIONS (Contd)

FEDERAL (Contd)

PPP-F-320	Fiberboard, Corrugated and solid sheet stock (Container Grade), and Cut Shapes
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Box, Fiberboard
PPP-B-640	Boxes, Fiberboard, Corrugated, Triple-Wall
PPP-S-760	Strapping, Nonmetallic, (and Connectors)

MILITARY

MIL-P-116	Preservation, Methods of
MIL-S-901	Shockproof Equipment, Class Hi (Hi-Impact), Shipboard Installation, Tests for
MIL-I-6869	Impregnants for Aluminum and Magnesium Alloy Castings
MIL-P-11268	Parts, Materials, and Processes Used in Electronic Communication Equipment
MIL-M-13231	Finishes for Ground Signal Equipment
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-I-46058	Coatings, Conformal (for Coating Printed Wire Assemblies)
MIL-P-55110	Printed Wiring Boards

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized Unit Loads, 40 in. X 48 in. 4 Way Partial and 4 Way Pallets
MIL-STD-210	Climatic Extremes for Military Equipment
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-276	Impregnation of Porous Non-Ferrous Metals
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-781	Reliability Tests; Exponential Distribution
MIL-STD-810	Environmental Test Methods

DRAWINGS

ELECTRONICS COMMAND

SC-DL-501092	Receiver-Transmitter RT-834( )/GRC
SC-DL-502716	Amplifier, Radio Frequency AM-3349( )/GRC-106
SC-GL-323149	Gage List

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(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer. Both the title and number or symbol should be stipulated when requesting copies).

3. REQUIREMENTS

3.1 First Article. - When specified in the contract or purchase order, the contractor shall furnish First Article units in accordance with 4.3.

3.2 Construction - Unless otherwise specified in the invitation for bid, physical construction of the equipment on order shall be in accordance with drawings listed below:

Equipment	Drawing and Data List
Receiver-Transmitter RT-834( )/GRC	SC-DL-501092
Amplifier, Radio Frequency AM-3349( )/ GRC-106	SC-DL-502716

3.3 Parts, materials, and processes: general - In addition to the requirements of this specification, the requirements of MIL-P-11268 including the selection requirements therein, shall apply.

3.3.1 Finish, protective (see 4.4) - The equipment shall be finished in accordance with MIL-F-14072. This includes finish of hardware, such as handles, screws, etc., and necessary touch-up after mounting. The final paint film on Type 1 surfaces shall be green color (Olive Drab), semi-gloss enamel matching a color chip provided by the procuring agency (see 6.3).

3.3.2 Marking (see 4.4) -

3.3.2.1 General - Marking shall conform to MIL-M-13231. Front panel marking shall be Group 1 as described in that specification.

3.3.2.2 Visibility - Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.3.2.3 Serial numbers - The following items shall have serial numbers:

Receiver-Transmitter, Radio RT-834( )/GRC

Amplifier, Radio Frequency AM-3349( )/GRC-106

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3.3.3 Impregnation of aluminum castings - All exterior aluminum castings shall be impregnated in accordance with MIL-STD-276, using Type II impregnating material in accordance with MIL-I-6869. Castings which have been impregnated shall be marked "IMP". The impregnation shall be used to correct general seepage only and shall not be used to correct poor foundry techniques or significant porosity (see 4.4).

3.3.4 Fungus - The equipment (including accessories) shall show no evidence of viable fungus or corrosion when subjected to the test specified in 4.11. Corrosion is any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish on metal surface.

3.3.5 Interchangeability - Like units, assemblies, subassemblies, modules, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment as determined by the tests specified in 4.9 and 4.10. However, electrical retuning and realignment of modules during production assembly operations may be expected and are permissible.

3.3.6 Printed wiring assemblies (see 4.4) - Printed wiring assemblies shall be processed and fabricated according to MIL-P-55110. Conformal coatings for the printed wiring assemblies, on printed circuit side only, shall be in accordance with MIL-I-46058, except that the thickness of coating shall be .003 $\pm$.002 inches.

3.4 Service conditions (see 4.7) - The equipment shall meet the following service conditions; there shall be no evidence of corrosion, damage or deterioration and where a test is referenced, meeting the test shall also be required.

3.4.1 Temperature (see 4.7.1) -

3.4.1.1 Low - During testing as specified in 4.7.1.1, the equipment shall meet the following requirements:

Paragraph	Measurement	Performance
3.6.3	IM Distortion	4.8.4
3.6.10	BFO	4.8.11
3.6.12	Noise Figure	4.8.13
3.6.21.4	Warm-up from cold start	4.8.23.4
3.6.21.5	Overall frequency stability	4.8.23.5
3.6.33.3	Power vs Temperature	4.8.35.3

After testing, the equipment shall meet full specification performance for these measurements.

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3.4.1.2 High - During the test of 4.7.1.2, the equipment shall meet the following requirements:

Paragraph	Measurement	Performance
3.6.3	IM Distortion	4.8.4
3.6.10	BFO	4.8.11
3.6.12	Noise Figure	4.8.13
3.6.21.3	Warm-up from hot oven	4.8.23.3
3.6.21.5	Overall frequency stability	4.8.23.5
3.6.33.3	Power vs Temperature	4.8.35.3

After testing, the equipment shall meet full specification performance for these measurements.

3.4.2 Humidity (moisture-resistance) - During the test of 4.7.2, the equipment shall meet the following requirements:

Paragraph	Measurement	Performance
3.6.3	IM Distortion	4.8.4
3.6.12	Noise Figure	4.8.13
3.6.21.5	Overall Frequency Stability	4.8.23.5
3.6.33.4	SSB Voice Power Output	4.8.35.4

After testing the equipment shall meet full specification performance for these measurements.

3.4.3 Elevation - The equipment shall meet full specification requirements for the following measurements both during and after testing as specified in 4.7.3:

Paragraph	Measurement	Performance
3.6.12	Noise Figure	4.8.13
3.6.33.2	SSB Voice Power output at 2.0000 MHz into dummy whip (4.8.1.4)	4.8.35.2

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3.4.4 Dust - After testing as specified in 4.7.4, the equipment shall meet full specification performance for the following measurements:

Paragraph	Measurement	Performance
3.6.12	Noise Figure	4.8.13
3.6.33.4	SSB Voice Power Output	4.8.35.4

3.4.5 Salt fog - After testing as specified in 4.7.5, the equipment shall show no evidence of corrosion as evidenced by any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish on metal surface and shall meet the following electrical requirements:

Paragraph	Measurement	Performance
3.6.12	Noise Figure	4.8.13
3.6.33.4	SSB Voice Power Output	4.8.35.4

3.4.6 Vibration, equipment - The amplitude of vibration of the mounted equipment (without shock mounts) shall not exceed two times the amplitude of the applied vibration at any frequency between 10 and 55 Hz (see 4.7.6). The equipment shall meet the following requirements:

Paragraph	Measurement	Performance
3.6.12	Noise Figure	4.8.13
3.6.21.6	Mobile Stability	4.8.23.6
3.6.33.4	SSB Power Output (2 tone)	4.8.35.4

3.4.6.1 Vibration, limited - When tested in accordance with 4.7.6.1, the following requirements shall be met:

Paragraph	Measurement	Performance
3.6.12	Noise Figure	4.8.13
3.6.33.6	CW Power Output	4.8.35.6

3.4.7 Bounce and shock - Units shall be capable of meeting the requirements of Table II.

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3.4.7 Bounce and shock - (contd) -

TABLE II - Bounce and Shock

Paragraph Heading	Paragraph Number	
Bounce	4.7.7	Specified performance (Note B) No physical damage
Shock; Bench handling	4.7.8	Specified performance (Note A) No physical damage
Shock; Ballistic	4.7.9	Specified performance (after shock test is completed). (Note B) except delete mobile stability and Phase Modulation and jitter. All physical damage shall be minor only.

NOTE A: Minor paint abrasion is permissible. After the test, the equipment shall meet the following requirements:

Noise Figure 4.8.13
SSB Power output 4.8.35.4

NOTE B: The equipment shall meet specified performance for the following tests:

Noise Figure 4.8.13
Mobile Stability 4.8.23.6
SSB Power output 4.8.35.4
Phase Modulation and Jitter 4.8.6 except that during this test in the receive mode, the audio distortion at the 10 mw output shall be not more than 3%. NOTE: The sequence of tests shall be as shown in Table II.

3.4.8 Acoustic noise - When measured as specified in 4.7.10, acoustic noise from rotating machinery, and mechanical vibration associated with normal operation on all classes of emission shall be no greater than 68 db above 10^{-16} watts/square centimeter (w/cm^2) (.0002 microbar) at low blower speed and 75 db above 10^{-16} w/cm^2 (.0002 microbar) at high blower speed.

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3.4.9 Immersion (see 4.7.11) - The receiver-transmitter or rf amplifier shall show no evidence of leakage after testing in accordance with 4.7.11.

3.4.10 Tuning knob stability - Friction on the "ANT TUNE" and "ANT LOAD" mechanisms shall be 70 ± 30 inch-ounce (see 4.7.12).

3.4.11 End stops - When tested as specified in 4.7.13, there shall be no permanent deformation of the end stops.

3.4.12 Electromagnetic compatibility (see 4.4) - The equipment shall meet the electromagnetic compatibility requirements of MIL-STD-461 (Table A - II-MIL-I-11748) and MIL-STD-462 for Class 1 equipments.

3.4.12.1 Test frequency range - The frequency range over which tests shall be conducted is from 14 KHz to 1000 MHz.

3.4.12.2 Harmonic emission - All harmonic emission shall be not less than 50 db below full power output.

3.4.12.3 Other extraneous outputs - Under receive conditions of the equipment, the total output at any antenna terminal on the equipment shall not exceed 200 micromicro-watts into a 50-ohm resistive load.

3.4.12.4 Parasitic oscillations - Parasitic oscillations shall not exceed a level of 80 db below the full power output in accordance with MIL-STD-461.

3.4.13 Air seal - When tested as specified in 4.7.14, the decrease in vacuum shall not exceed 0.01 pounds per square inch, gauge (PSIG), per minute.

3.5 Preconditioning (see 4.6) - All equipments, including First Article Samples, produced under this specification shall be preconditioned in accordance with the following, prior to submission to the Government for acceptance, preproduction testing or reliability testing.

3.5.1 Bounce preconditioning - The equipment shall be capable of meeting the requirements specified herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6.1 (see also 4.5).

3.5.2 Burn-in/debugging - Each equipment on order shall be subjected to a burn-in/debugging of 48 hours. The last 12 hours of this 48 hours of burn-in/debugging shall be failure free. Each equipment shall meet the noise figure (3.6.12) and the SSB voice power output (3.6.33.4) requirements, after successfully completing the 48 hour burn-in/debugging of 4.6.2.

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3.6 Electrical requirements (see 4.8) - The equipment shall meet all the requirements contained in this specification. Each requirement shall apply equally to all frequencies within the frequency range. Specific designations of test frequencies in the specification or contractor's prepared test procedures shall be construed as examples to show minimum number of frequencies usually selected to determine compliance and shall not be construed as restricting testing at other than the listed frequencies. Testing shall be performed on designated frequencies plus any other frequency which is not designated if such additional testing is deemed necessary to determine and assure full compliance across the frequency range. Failures encountered during test on any other frequency than those specifically designated shall also be cause for rejection of the unit (or lot when sampling inspection is used).

3.6.1 Warm-up (see 4.8.2) - After the radio set has been in standby position for 5 minutes and within 60 seconds after the set has been switched to operate, the noise figure shall not be poorer than 12 db and the SSB voice power output (3.6.33.4) shall be not less than 400 watts PEP.

3.6.2 Frequency response - When tested as specified in 4.8.3 and 4.8.14, the overall frequency response of the receiver-transmitter measured between antenna and audio shall conform to the following:

Audio Frequency	Response
300 and 3500 Hz	End points not more than 3 db down
400 to 3400 Hz	Peak to valley ratio equal to or less than 2.5 db
50 and 3850 Hz	At least 10 db below reference (1000 Hz)
25 and 4000 Hz	At least 40 db down below reference (1000 Hz)
Beyond 4350 Hz above and below the carrier frequency	At least 50 db down below reference (1000 Hz)

Once the overall response curve falls below 3.5 db, there shall be no reversal in the slope of the curve until the response drops below 60 db. The response,in the range of 0 Hz (carrier frequency) to 4,350 Hz in the opposite sideband, shall be governed by the requirements of 3.6.7.

3.6.3 Intermodulation distortion - When measured as specified in 4.8.4, intermodulation distortion in the output of the radio set shall have a value of more than 35 db below either of the two equal output frequencies (4.8.1.6) corresponding to two audio input modulating tones with frequencies of 1,500 and 2,500 Hz when the radio set is operating at rated peak envelope power.

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3.6.4 Hum and noise - When measured as specified in 4.8.5, hum and noise output, measured at the antenna terminals of the radio set, shall be not less than 50 db below the transmitted output level of either of two equal audio tones (4.8.1.6) modulating the radio set to rated peak envelope power output in the SSB mode.

3.6.5 Phase modulation and jitter - When tested as specified in 4.8.6, there shall be no detectable undesired phase modulation or jitter in any mode of operation under any normal combination of service conditions at any frequency in the range of 2.0000 to 29.9999 MHz.

3.6.6 Carrier suppression - When measured as specified in 4.8.7, the carrier shall be suppressed a minimum of 50 db below the level of either tone specified in 4.8.1.6 when SSB operation is employed.

3.6.7 Opposite sideband suppression (see 4.8.8) - In SSB operation, suppression of all components in the channel of the opposite sideband shall be not less than 50 db below either of the two desired sideband test tones of 4.8.1.6, except for intermodulation products, for all frequencies within the intelligence bandwidth of the radio set at all output levels up to and including rated power output. The identification of I.M. products in the opposite sideband will be accomplished by removal of one of the test tones of 4.8.1.6.

3.6.8 CW keying -

(a) Reception: When tested as specified in 4.8.9(a), CW telegraph signals up to 25 dot cycles per second shall be received without more than 10 percent distortion.

(b) Transmission (see 4.8.9(b) - Keying speeds up to 25 dot cycles per second shall be possible. The CW frequency shall be 2000 $\pm$ 6 Hz above the carrier frequency (fo). The radiated signal shall fall within the shaded portion of Figure 1.

(c) CW attack time - When the transmitter is keyed-on in the CW mode, the power output shall be at least 90% of full power output within 30 milliseconds.

3.6.9 Signal level meter - When tested as specified in 4.8.10, the signal level meter shall give an appropriate indication of signal level of the incoming signal in db above automatic gain control (agc) threshold; keyed, it shall indicate the rf output.

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3.6.10 BFO (see 4.8.11) - Using the beat frequency oscillator (bfo) control, it shall be possible to vary the resultant audio output frequency at least 3.5 KHz, but not more than 5.5 KHz from each side of zero beat with an incoming signal of f_0 + 2 KHz as the ambient temperature around the receiver is changed from -40 to +65 degrees Centigrade.

3.6.11 Overall gain (see 4.8.12) - With a signal of 4 microvolts (uv) fed into the antenna input, and using a 75-ohm source, the audio output power shall be 2.0w with the agc on. Change in over-all gain over the tuning range shall not exceed a ratio of 4 to 1. Change in over-all gain at any frequency in the range of 2.0000 to 29.9999 MHz under any condition of change in humidity, temperature, altitude, or as a result of any mechanical test specified herein, shall not exceed 3 db.

3.6.12 Noise figure - The noise figure of the radio set, when measured as specified in 4.8.13, shall not be poorer than 12 db.

3.6.13 Tuned circuit mistracking - Mistracking of the RF Amplifier (Module 1A12), at any frequency shall not exceed 4 db (see 4.8.15).

3.6.14 Desensitization - Desensitization, when measured in accordance with 4.8.16, shall comply with the limits specified for cross modulation (3.6.25).

3.6.15 IF bandwidth (see 4.8.17) - The intermediate-frequency (IF) bandwidth characteristics shall be as shown in Figure II. The response curve shall fall in the shaded area and once the response reached -60 db, it shall remain at least -60 db.

3.6.16 IF output (see 4.8.18) - With a 4 uv single tone signal input through a 75 ohm dummy antenna, the IF output voltage across 50 ohms shall be not less than 20 Milli-volts (MV). The receiver shall be capable of this performance with the service selector and agc switch in any operating position except that this test shall not be performed with the service selector switch in the (AM) position. The IF output frequency shall nominally be 1.752 MHz.

3.6.17 Squelch - When tested in accordance with 4.8.19, the receiver must squelch within 5 seconds.

3.6.18 AGC and blocking (see 4.8.20) - The automatic gain control shall be such that the audio power of the receiver-transmitter unit will increase not more than 4 db when the signal input is increased from 5 to 1000 microvolts and not more than 8 db when the signal input is increased from 5 microvolts to 1.0 volt. The output shall not decrease with increasing signal input from 5 microvolts to 1.0 volts. These requirements shall apply with any type of emission or mode of detection referenced herein. The agc hang time shall be 900 $\pm$ 200 milliseconds (ms). The agc attack time shall be 30 $\pm$ 15 ms.

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3.6.19 Frequency standard adjustment and calibration (see 4.8.21) - Following a minimum aging period of 168 hours, the adjustment shall provide a warping range of at least ± 10 Hz from the standard frequency and a total warping range of at least 110 Hz. The frequency adjustment control shall then be precisely aligned to within 0.1 Hz of 5.0 MHz.

3.6.20 Internal and external standards (see 4.8.22) - The internal standard output shall provide 0.2 volt rms of signal across 50 ohms. The set shall operate properly when using an external standard supplying signal levels from 75 mv to 3 volts rms. The impedance of the connector in this application shall be 50 ohms nominal.

3.6.21 Frequency stability (see 4.8.23) - After 1 hour of warm-up, except where otherwise noted, the frequency stability and accuracy of the crystal standard oscillator shall meet the requirements stated below over the entire operating frequency range, and under any combination of extreme operating and cold storage conditions stated therein.

3.6.21.1 Long term stability - After a one hour warm-up period, the radio set shall be operated in the receive mode for a total of 168 hours. After the total accumulation of 168 hours has been reached, the oscillator circuit and aging rate of the crystal shall be such that the long term stability of the frequency standard is 1 part in 10^7 per week (see 4.8.23.1).

3.6.21.2 Short term stability - The equipment's short term stability shall be 5 parts in 10^7 when measured over a time duration of from one second to 1 hour over the entire operating ambient temperature range (see 4.8.23.2).

3.6.21.3 Warmup from hot oven (see 4.8.23.3) - With the oven on for a period of one hour, the frequency standard shall meet the requirements of 3.6.21.2 within 5 seconds after oscillator power is supplied.

3.6.21.4 Warm-up from cold start (see 4.8.23.4) - With the oven off and the oven temperature at external ambient temperature, the frequency standard shall meet the requirements of 3.6.21.1 and 3.6.21.2 within 1 hour after oven and oscillator power has been applied. After the first 12 minutes of warm-up, the stability shall be within 1 part of 10^6 of the frequency obtained after one hour of warm-up.

3.6.21.5 Over-all frequency stability (see 4.8.23.5) - The tuned circuits and the 100 Hz lock circuits shall be such that the stability of the transmitted output and received input signals (except for drift due to the signal being received) shall be the same as that specified for the frequency standard as indicated in 3.6.21.1 when the radio set is detented at a 100 Hz point. When the radio set is unlocked, the over-all frequency drift shall be not more than 100 Hz total. Frequency drift for periods less than 1 hour shall be a proportional part of the drift for one hour and no sudden change in frequency shall occur in excess of one tenth of a hertz per second.

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3.6.21.6 Mobile stability (see 4.8.23.6) - When subjected to the equipment tests of (4.7.7) and (4.7.8) and with the radio set in a 100 Hz detent position, the frequency change of the transmitted or received frequency shall not exceed 5 Hz. When subjected to the bounce and vibration tests with the radio set unlocked, the frequency change of the received signal shall not exceed 15 Hz. The requirements of 3.6.21.5 shall be met.

3.6.21.7 Stability vs line voltage - When tested as specified in 4.8.23.7, frequency change from the nominal shall be not more than specified in 3.6.21.2.

3.6.22 Frequency accuracy - When measured in accordance with 4.8.24, the carrier frequency shall be locked to the internal standard and shall not differ from the dial setting by more than the following: (Difference between the internal standard frequency and 5 MHz) times (dial setting in MHz divided by 5 MHz).

3.6.23 Vernier tuning (see 4.8.25) - The frequency variation shall be not less than ± 500 Hz nor more than ± 700 Hz with approximately 360° of rotation of the vernier tuning knob. In a given mechanically fixed (OFF) position, the vernier tuning shall be disabled and the receiver shall be locked at the 100 Hz detent point. While the receiver is being vernier tuned, the transmitter output frequency shall not vary from the previously set 100 Hz detent point.

3.6.24 Internal signals (see 4.8.26) - An internal signal shall be defined as a signal generated within the receiver-transmitter which is capable of being detected by the equipment. Shielding and filtering within the receiver-transmitter shall be to the extent required to reduce the level of internal signals equal to or below the equivalent signal level of one uv referred to antenna input.

3.6.25 Cross-modulation - Cross-modulation response when measured in accordance with 4.8.27 shall comply with the following limits:

Level of undesired signal	Maximum separation between desired and undesired signals in percent of desired signal
0.01 volts	0.6
0.1 volts	2.5
1.0 volts	4.5
6.0 volts	7.0

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3.6.26 IF rejection (see 4.8.28) - The rejection ratio shall be not less than 80 db, except that with the receiver set at 22.9990 MHz, a desired signal of 23.000 MHz, and an undesired signal of 20.5000 MHz, the required rejection shall be not less than 50 db.

3.6.27 Image rejection (see 4.8.29) - The rejection ratio shall be not less than 80 db.

3.6.28 Cross-over rejection (see 4.8.30) - Throughout the range of the receiver, rejection of spurious interference of the cross-over type shall be not less than 60 db, except for the following:

Receiver Freq (MHz)	Reference Signal (MHz)	Undesired Signal (MHz)	Req'd Rejection (db)
2.9200	2.9210	2.9158	40
2.9200	2.9210	2.9173	40
3.3030	3.3040	3.3007	40
5.8020	5.8030	5.7993	40
5.8020	5.8030	5.8005	40
8.6270	8.6280	8.6240	40
8.6270	8.6280	8.6256	40
11.3340	11.3350	11.3338	45
11.3340	11.3350	11.3325	45
21.2540	21.2550	21.2450	50
21.2540	21.2550	21.2517	40

3.6.29 Receiver-SSB intermodulation (see 4.8.31) - When a composite test signal consisting of two equal sideband voltages, differing in frequency from the carrier by between 300 to 3,500 Hz, and from each other by not more than 500 Hz is applied to the input terminals of the receiver, any resulting distortion component in the output signal shall be not less than 40 db below the output level of the sideband voltage of each tone when the total output level is set at one milliwatt (mw) and 35 db below the output level when the output is set at 10 mw.

3.6.30 BFO and carrier leakage - When measured as specified in 4.8.32, the IF output circuit shall be so isolated that the BFO leakage shall be not less than 35 db below the signal level and the carrier leakage shall be not less than 40 db below the signal level when measured at the IF output connection.

3.6.31 Audio distortion (see 4.8.33) - Total audio distortion shall be no greater than 5 percent with 2.0 w power output and not greater than 1 percent at 10 mw audio power output.

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3.6.32 Audio power (see 4.8.34) - By means of the front panel control, the audio output shall be adjustable from a nominal zero to not less than 2.0 w working into a nominal 600-ohm load.

3.6.33 RF power output (see 4.8.35) - The radio set shall have the following levels of power output over the frequency range of 2.0000 to 29.9999 MHz for the various modes of emission when powered from a 24-volt vehicular generating system:

3.6.33.1 Power vs frequency - When tested as specified in 4.8.35.1, power output shall conform to 3.6.33.4, 3.6.33.5 and 3.6.33.6.

3.6.33.2 Antenna tuner - When tested as specified in 4.8.35.2, power output into a dummy whip load shall not be less than the following:

Frequency (MHz)	RF PEP Volts/50 ohms	Frequency (MHz)	RF PEP Volts/50 ohms	Frequency (MHz)	RF PEP Volts/50 ohms	Frequency (MHz)	RF PEP Volts/50 ohms
2.0000	65	8.5000	110	16.9900	120	25.0000	125
2.2500	70	8.9900	110	17.0000	120	25.5000	125
2.4900	75	9.0000	110	17.5000	120	25.9900	120
2.5000	65	9.5000	110	17.9900	120	26.0000	125
2.7500	60	9.9900	115	18.0000	120	26.5000	125
2.9900	60	10.0000	105	18.5000	110	26.9900	125
3.0000	55	10.5000	105	18.9900	115	27.0000	105
3.2500	55	10.9900	110	19.0000	125	27.5000	105
3.4900	50	11.0000	120	19.5000	125	27.9900	105
3.5000	50	11.5000	120	19.9900	125	28.0000	105
3.7500	50	11.9900	120	20.0000	120	28.5000	105
3.9900	45	12.0000	115	20.5000	120	28.9900	105
4.0000	45	12.5000	115	20.9900	120	29.0000	105
4.5000	40	12.9900	115	21.0000	110	29.5000	105
4.9900	40	13.0000	120	21.5000	110	29.9900	105
5.0000	90	13.5000	120	21.9900	110		
5.5000	85	13.9900	120	22.0000	120		
5.9900	85	14.0000	105	22.5000	120		
6.0000	75	14.5000	110	22.9900	120		
6.5000	75	14.9900	110	23.0000	125		
6.9900	75	15.0000	120	23.5000	125		
7.0000	75	15.5000	120	23.9900	125		
7.5000	75	15.9900	120	24.0000	125		
7.9900	70	16.0000	120	24.5000	125		
8.0000	110	16.5000	120	24.9900	125		

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3.6.33.2 Antenna tuner - (contd) -
NOTE: When using a doublet antenna for the asterisked frequencies listed in 4.6.2, the tuner shall be capable of matching a doublet antenna with a standing wave ratio of not more than 2.5 to 1.

3.6.33.3 Power vs temperature - When tested as specified in 4.8.35.3, power output shall conform to 3.6.33.2.

3.6.33.4 SSB power output (see 4.8.35.4) - When modulated by the standard test message of 4.8.1.7, the radio set shall be capable of a peak envelope power output of at least 400 watts. When modulated by two equal sinusoidal audio tones (1500 and 2500 Hz), the peak envelope power output shall be 400 watts + 2, to -1 db.

3.6.33.5 Compatible AM (see 4.8.35.5) - The peak envelope power output of the compatible AM signal (carrier and upper sideband) when modulated by the standard test message of 4.8.1.7 shall be at least 400 watts. The carrier power shall be 70 watts + 2 to -1 db.

3.6.33.6 CW and FSK (see 4.8.35.6) - The average power output shall be 200 watts +2 to -1 db.

3.6.33.7 Power vs line voltage - When tested as specified in 4.8.35.7, the PEP on two-tone shall be not less than 400 w + 2, -1 db from 25 to 30 volts, at any of the specified frequencies. There shall be normal operation in all respects, except reduced power is permissible, between 22 and 24 vdc. Between 24 and 25 volts, the PEP on two-tone shall be not less than 400 w -1.5 db.

3.6.34 Interlock (see 4.8.36) - The interlock shall automatically disconnect power control circuits when the Amplifier AM-3349 is extracted from its case. This disconnect function shall take effect when the unit is unfastened from its case, and pulled forward a maximum of one-half inch. When the equipment is returned to its case, the equipment shall operate.

3.6.35 PA overload - When tested as specified in 4.8.37, the power amplifier (AM-3349) circuit shall be de-energized within 200 $\pm$ 100 milliseconds after application of 25 percent and 200 percent overload. The overload control shall not be affected by momentary surges. The reset shall operate manually.

3.6.36 Reverse current - When tested as specified in 4.8.38, there shall be no evidence of damage or abnormal operation.

3.6.37 Oven protection - When tested as specified in 4.8.39, the protection circuit shall operate properly.

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3.6.38 Antenna transfer - When tested as specified in 4.8.40, the antenna circuit shall be transferred from the receiver to the PA output as in normal operation. The flag switch shall operate properly.

3.6.39 Power drain - When tested as specified in 4.8.41, the total power drain shall not exceed 1161 watts for 80 of the 90 test frequencies and in no case shall the power drain exceed 1215 watts.

3.6.40 VOX operation - When tested as specified in 4.8.42, all operations during the test shall be normal.

3.6.41 Metering - When tested as specified in 4.8.43, all meter indications shall be within the proper spaces.

3.6.42 Tune power (see 4.8.44) - With the radio set in the TUNE condition and no modulation applied, the TUNE power output shall be 70 w + 2, to -1 db.

3.6.43 CW sidetone - The test of 4.8.45 shall show proper operation and limits.

3.6.44 Front end protection (see 4.8.46) - The front end protection circuit shall operate when the rf input to the RT-834/GRC reaches 8 ± 1 volts. When the excessive voltage is removed, the radio set shall function normally with no degradation of performance. The circuit shall function correctly at any frequency in the 2.0 to 30.0 MHz range of the radio set.

3.7 Operational inspection - The operational test specified in 4.13 shall be performed to determine that the radio set is operating satisfactorily prior to packaging.

3.8 Technical literature, tools, and running spare parts - Technical literature, tools, and running spare parts shall be furnished if specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order (see 6.2(h)).

3.9 Workmanship - The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs:

(a) In MIL-P-11268:

General requirements for plastic material and plastic parts.

Wiring and cabling, including:

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3.9 Workmanship - (contd) -

Slack
Protection
Clearance
Splicing
Connections, general
Grounding, general
Shielding on wire and cable
Soldering and brazing

General requirements for securing of parts
Cleaning of units
Process for welding
Cleaning after welding
Controls
Self-tapping screws
Cleaning prior to welding

(b) In this specification:

3.2 Construction
3.3.1 Finish
3.3.2 Marking
3.3.6 Printed Wiring Assemblies

3.10 Reliability test (see 4.16) - Each radio set undergoing reliability testing shall demonstrate a specified mean-time-between-failures (MTBF) of 125 hours. The required reliability shall be demonstrated by a reliability test using the Reliability Test Plan (Appendix A).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspection - Inspection shall be classified as follows:

a. First Article inspection (does not include preparation for delivery) (see 4.3).

b. Inspection covered by subsidiary documents (see 4.4).

c. Quality conformance inspections.

(1) Quality conformance inspection of equipment before preparation for delivery (see 4.5).

(2) Quality conformance inspection of preparation for delivery (see 4.14).

4.3 First Article - Unless otherwise specified in the contract or purchase order, the First Article inspection shall be performed by the contractor.

4.3.1 First Article units - The contractor shall furnish six (6) First Article units of Radio Set AN/GRC-106( ).

4.3.2 First Article inspection - The First Article inspection shall consist of the inspections specified in subsidiary documents covering the items listed in 4.4, preconditioning specified in 4.6, and the inspections specified for Group A, Group B, Group C and Group D (see Table IV, V, VI and VII respectively). The inspection shall be performed in the following order: 4.4, 4.6, Group A and Group B for all units and designated units for specific Group C and Group D tests in the order shown in Table III with the balance of Group C tests performed on all First Article units.

TABLE III

Inspection (Note 1)	Unit 1	Unit 2	Unit 3	Units 4-6
Vibration, internal		1		
Vibration, equipment	1			
Temperature		2		
Bounce	2			
Shock, bench handling	3			
Humidity	4			
Fungus (Note 2)		3		
Salt fog (Note 2)			1	
Shock ballistic			2	
Immersion			3	
Reliability test				1

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4.3.2 First Article inspection - (contd) -

NOTE 1: Other First Article inspection may precede, follow, or be interspersed between the tests.

NOTE 2: The equipment shall be thoroughly washed, cleaned, dried, and refurbished, if necessary, before proceeding with subsequent tests.

4.4 Inspection covered by subsidiary documents - The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification.

Item	Where required
Finish	3.3.1
Marking	3.3.2
Printed wiring	3.3.6
Electromagnetic compatibility	3.4.12
Casting impregnation	3.3.3
Workmanship	3.9

4.5 Quality conformance inspection of equipment before preparation for delivery - Each unit of each lot of equipment shall be preconditioned (see 3.5) prior to inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.5. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. Failure to meet any requirement as a result of preconditioning shall be classified as a defect and recorded as such. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government, at its discretion, may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements (see 6.10). Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection - Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table IV. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection utilizing the general inspection levels, and AQLs indicated in Table IV.

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4.5.1 Group A inspection - (contd) -

TABLE IV - Group A Inspection

Inspection	Module Reference	Req Para	Insp Para	AQL Major	AQL Minor
Visual & Mechanical		3.9	4.12		
RT-834 Modules					
Chassis	1A1			0.65%	2.5%
100 Hz Synthesizer	1A1A1			0.65%	2.5%
100 KHz Synthesizer	1A2			0.65%	2.5%
Frequency Standard	1A3			0.65%	2.5%
10 & 1 KHz Synthesizer	1A4			0.65%	2.5%
Transmit IF & Audio	1A5			0.65%	2.5%
Frequency Divider	1A6			0.65%	2.5%
Receive IF	1A7			0.65%	2.5%
Translator	1A8			0.65%	2.5%
MHz Synthesizer	1A9			0.65%	2.5%
Receive Audio	1A10			0.65%	2.5%
DC to DC Converter	1A11			0.65%	2.5%
RF Amplifier	1A12			0.65%	2.5%
AM-3349 Modules					
Chassis & Plenum	2A1			0.65%	2.5%
Turret	2A2			0.65%	2.5%
Antenna Coupler	2A3			0.65%	2.5%
Discriminator	2A4			0.65%	2.5%
Front Panel	2A5			0.65%	2.5%
Case, Blower and Converter	2A6			0.65%	2.5%
Relays	2A7			0.65%	2.5%
Driver	2A8			0.65%	2.5%
P.A. Stator	2A9			0.65%	2.5%
Flag Switch				0.65%	2.5%
Assemblies, printed Circuit Boards				0.65%	2.5%
Assemblies Receiver-Transmitter RT-834				1.0%	4.0%
Assembled Power Amplifier AM-3349				1.0%	4.0%

(contd)

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4.5.1 Group A inspection - (contd) -

TABLE IV - Group A Inspection - (contd)

Inspection	Req Para	Insp Para	AQL Major	AQL Minor
Electrical				
RT-834				
Noise Figure	3.6.12	4.8.13)	* 1.0 DPHU for the group	N/A
Squelch	3.6.17	4.8.19)		
Frequency accuracy	3.6.22	4.8.24)		
Vernier tuning	3.6.23	4.8.25)		
Audio distortion	3.6.31	4.8.33)		
Audio power	3.6.32	4.8.34)		
Vox operation	3.6.40	4.8.42)		
AM-3349( )/GRC-106				
Interlock	3.6.34	4.8.36)	*1.0 DPHU for the group	N/A
Antenna transfer	3.6.38	4.8.40)		
Electrical				
AN/GRC-106				
Warm-up	3.6.1	4.8.2)		
Intermodulation distortion	3.6.3	4.8.4)		
Hum and noise	3.6.4	4.8.5)		
Carrier suppression	3.6.6	4.8.7)		
Opposite sideband suppression	3.6.7	4.8.8)	*1.0 DPHU for the group	
SSB power output	3.6.33.4	4.8.35.4)		
Compatible am power output	3.6.33.5	4.8.35.5)		
CW & FSK power output	3.6.33.6	4.8.35.6)		
Metering	3.6.41	4.8.43)		
Tune power output	3.6.42	4.8.44)		
Power vs frequency	3.6.33.1	4.8.35.1	1.0	N/A
Freq. Std. Adjustment & calibration	3.6.19	4.8.21	1.0	N/A
Tuning knob stability	3.4.10	4.7.12	1.0	N/A
Air seal	3.4.13	4.7.14	1.0	N/A
Operational	3.7	4.13	1.0	N/A

* Defects per hundred units

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4.5.1.1 Order of inspection within Group A - Group A inspection shall be performed in any order acceptable to the Government except that the air seal test (4.7.15) shall precede the operational inspection (4.13) which shall be last (see 6.7), immediately prior to packaging.

4.5.2 Group B inspection - Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units which have been subjected to and met Group A inspection. This inspection shall conform to Table V and to the special inspection level of Table I of MIL-STD-105.

TABLE V - Group B Inspection

Inspection	Req Para	Insp Para	AQL (see 4.5.2.1)
Interchangeability			
Mechanical	3.3.5	4.9	6.5%
Electrical	3.3.5	4.10	2.5%
Immersion	3.4.9	4.7.11	4.0%
Signal level meter	3.6.9	4.8.10	4% for the group
IF Bandwidth	3.6.15	4.8.17	
IF Output	3.6.16	4.8.18	
AGC and blocking	3.6.18	4.8.20	
IF rejection	3.6.26	4.8.28	
Image rejection	3.6.27	4.8.29	
BFO & carrier leakage	3.6.30	4.8.32	
CW sidetone	3.6.43	4.8.45	4% for the group
Overall gain	3.6.11	4.8.12	
Desensitization	3.6.14	4.8.16	
Internal and External standards	3.6.20	4.8.22	
Cross modulation	3.6.25	4.8.27	
Reverse current	3.6.36	4.8.38	4% for the group
Oven protection	3.6.37	4.8.39	
Power drain	3.6.39	4.8.41	
Front end protection	3.6.44	4.8.46	

4.5.2.1 Group B sampling plan - The Acceptable Quality Level (AQL) shall be as specified in Table V, and the inspection level shall be S-4. All defects shall be considered major.

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4.5.2.2 Order of inspection within Group B - Group B inspection shall be performed in any order which is satisfactory to the Government except that the immersion test shall be last.

4.5.3 Group C inspection - This inspection shall consist of the tests specified in Table VI, and shall normally be performed on sample units that have been subjected to and met Group A and Group B inspection (see 6.6.1).

TABLE VI - Group C Inspection

Inspection	Req Para	Insp Para
Sub-group I (see 4.5.3.1.1)		
Tuned circuit mistracking	3.6.13	4.8.15
Power vs Line Voltage	3.6.33.7	4.8.35.7
P.A. Overload	3.6.35	4.8.37
Sub-group II (see 4.5.3.1.2)		
Temperature	3.4.1	4.7.1
CW Keying	3.6.8	4.8.9
Limited Vibration	3.4.6.1	4.7.6.1
Frequency response (Transmitter)	3.6.2	4.8.3
Frequency response (Receiver)	3.6.2	4.8.3
Internal Signals	3.6.24	4.8.26
Cross over rejection	3.6.28	4.8.30
Receiver SSB intermodulation	3.6.29	4.8.31
Bounce	3.4.7	4.7.7
Phase modulation and jitter	3.6.5	4.8.6
Warmup from cold start	3.6.21.4	4.8.23.4
Mobile stability	3.6.21.6	4.8.23.6
Power vs temperature	3.6.33.3	4.8.35.3
BFO	3.6.10	4.8.11
Sub-group III (see 4.5.3.1.3)		
Humidity	3.4.2	4.7.2
Vibration, equipment	3.4.6	4.7.6
Shock ballistic	3.4.7	4.7.9
Acoustic noise	3.4.8	4.7.10
End stops	3.4.11	4.7.13
Freq. stability	3.6.21	4.8.23
Antenna Tuner	3.6.33.2	4.8.35.2
		(cor td)

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4.5.3 Group C inspection - (contd) -

TABLE VI - Group C Inspection - (contd)

Inspection	Req Para	Insp Para
Sub-group IV (see 4.5.3.1.4)		
Fungus	3.3.4	4.11
Elevation	3.4.3	4.7.3
Dust	3.4.4	4.7.4
Salt Fog	3.4.5	4.7.5
Shock, bench handling	3.4.7	4.7.8
Electromagnetic Compatibility	3.4.12	4.4

4.5.3.1 Sampling for Group C inspection - Units selected for each Group C inspection shall be selected without regard to their quality in accordance with the following:

4.5.3.1.1 Subgroup I - For this subgroup, two units from the first 25 production units shall be selected for the first Group C inspection. For subsequent Group C inspections, two units from each successive 100, or two per month, if less than 100 are produced monthly, shall be selected. For this subgroup, the maximum number of units to be tested in a single month shall be 8.

4.5.3.1.2 Subgroup II - For this subgroup, two units from the first 25 production units shall be selected for the first Group C inspection. For subsequent Group C inspections, two units per month shall be selected.

4.5.3.1.3 Subgroup III - For this subgroup, two units from the first 25 production units shall be selected for the first Group C inspection. For subsequent Group C inspections, two units every four (4) months shall be selected.

4.5.3.1.4 Subgroup IV - For this subgroup, two units from the first 25 production units shall be selected for the first Group C inspection. No subsequent Group C inspection for the tests in this subgroup will be required.

4.5.3.2 Order of inspection within Group C - Group C inspection shall be performed in any order acceptable to the Government.

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4.5.3.3 Noncompliance - The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and, (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract, through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government), and Groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until Group C reinspection results have shown that the corrective action was effective.

4.5.4 Group D inspection - This inspection shall consist of the tests specified in Table VII, and shall be performed on units from lots which have been subjected to and met Group A and Group B inspection.

TABLE VII - Group D Inspection

Inspection	Req Para	Insp Para
Reliability Test	3.10	4.16

4.5.5 Reinspection of conforming Group B, Group C and Group D sample units - Unless otherwise specified, sample units which have been subjected to and passed Group B, Group C and Group D inspection may be accepted on contract, provided that they are resubjected to and pass Group A reinspection after repair of all visible and known damage.

4.5.5.1 Sampling for reliability testing of equipment - Randomly selected samples shall be in accordance with paragraph 40.0 of Appendix A.

4.5.5.2 Reliability testing - For each production lot, delivery of contract item shall not be made pending successful completion of the reliability testing.

4.6 Preconditioning - The preconditioning tests of paragraphs 4.6.1 and 4.6.2 below shall precede all other electrical tests of the equipment. Bounce preconditioning shall precede burn-in/debugging.

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4.6.1 Bounce preconditioning - The units, with shock mounts (if any) removed or blocked, shall be placed in its normal operation position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, aligned, cleaned, or otherwise changed prior to subjection to inspection (see 3.5.1).

4.6.2 Burn-in/debugging -

4.6.2.1 Test conditions (see 3.5.2) - The equipment shall be operated for a period of 48 hours, with the following temperature and vibration conditions:

Temperature: + 40°C ± 5°C

Vibration: 2.2G ± 10% peak acceleration at any non-resonant frequency between 20 and 55 Hz measured on the Mounting MT-3140/GRC-106. The duration of the vibration shall be at least 10 minutes during each one hour of equipment operating time. SSB voice power output (3.6.33.4) using a 50 load (4.8.1.3) and noise figure (3.6.12) shall be measured at the beginning and end of the 48 hour burn-in/debugging at the following frequencies:

*2.0000 MHz	*8.6666 MHz	16.0000 MHz	23.0000 MHz
2.8300 MHz	9.0000 MHz	17.2000 MHz	24.9000 MHz
*3.1111 MHz	*10.0000 MHz	18.3000 MHz	25.0000 MHz
3.8300 MHz	11.4000 MHz	19.5000 MHz	26.0000 MHz
*4.2222 MHz	12.7000 MHz	20.6000 MHz	27.0000 MHz
*5.3333 MHz	13.1000 MHz	*21.8888 MHz	28.0000 MHz
*6.4444 MHz	14.8000 MHz	22.0000 MHz	*29.9999 MHz
*7.5555 MHz	15.0000 MHz		

After initial measurements have been made, the radio set shall be tuned to 2.0000 MHz and operated with a transmit to receive ratio of one hour transmit to one hour receive. After 12 hours of the above conditions, SSB voice power output shall be measured at all of the above frequencies.

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4.6.2.1 Test conditions (see 3.5.2) - (contd) -
The transmit-receive cycling shall then continue for an additional 12 hours and the power output test repeated. The above test shall be repeated and recorded every 12 hours for the duration of the burn-in/debugging. During the receive portion of each transmit-receive cycle, a functional receiver check shall be made as follows: A frequency of 2.0020 MHz shall be fed into the antenna input of the RT-834, and the resulting audio output continuously monitored.

4.6.2.2 Failures - If the equipment fails any test during burn-in/debugging, the contractor shall perform all necessary rework to correct the failure cause.

If the failure occurs during the first 36 hours of burn-in/debugging, the failure will be corrected and the equipment returned to burn-in to complete the 48 hour period.

If the failure occurs during the last 12 hours of the burn-in/debugging period, the equipment shall be returned to burn-in/debugging until it successfully completes a 12 hour failure free period.

Equipment failures during Group A, B or C testing will be corrected and subjected to burn-in/debugging until it successfully completes a 12 hour failure free period.

4.6.2.3 Compliance - Units passing burn-in/debugging will be submitted to Group A and Group B inspection, in that order, and Group C or Group D, as specified herein, without benefit of repair, cleaning, replacement, or other changes unless otherwise specified herein.

4.7 Service conditions tests (see 3.4) -

4.7.1 Temperature test (see 3.4.1) -

4.7.1.1 Low temperature - The equipment shall be tested in accordance with Method 502, procedure 1, of MIL-STD-810. The storage temperature shall be -80°C for 24 hours, or until practical thermal equilibrium is reached, and the operating temperature shall be -40°C. Prior to testing, the equipment shall meet full specification requirements for measurements of 3.4.1.1.

a. Upon reaching step 5 of above procedure, the equipment shall be turned on and the warmup from cold start test (4.8.23.4) shall be performed. The equipment shall then be operated, key down, for a period of one (1) hour. Immediately following the one hour key down, the equipment shall be subjected to all other tests listed in 3.4.1.1.

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4.7.1.1 Low temperature - (contd) -

b. In step 6, the equipment shall be kept in standby until stabilized.

c. After stabilization at standard ambient conditions, perform all tests specified in 3.4.1.1.

4.7.1.2 High temperature - The equipment shall be tested in accordance with Method 501, Procedure 1, of MIL-STD-810. The operating temperature shall be + 65° centigrade. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.4.1.2.

a. On reaching step 5, all measurements specified in 3.5.1.2 shall be made. After taking these measurements, the set shall be operated key down in the SSB/NSK mode, with two-tone input (4.8.1.6) into a dummy whip (4.8.1.4) for 4 hours at 2.0000 MHz. At the end of the first, second, and third hours, intermodulation distortion (4.8.4) and power output (4.8.35.3) shall be measured. At the end of the fourth hour, all measurements specified in 3.4.1.2, except warmup from hot oven, shall be made.

b. In step 6, the equipment shall be kept in Standby until stabilized.

c. After stabilization at standard ambient, perform all tests in 3.4.1.2.

4.7.2 Humidity (moisture-resistance) test (see 3.4.2) - The equipment shall be tested in accordance with Method 507, Procedure II, of MIL-STD-810. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.4.2. Measurements shall be made during the periods shown on the figure for the cycle. Do not move the equipment from the humidity chamber for measurements, and after measurements have been completed, the service selector switch shall be placed in the "Oven On" position. If any equipment fails to meet the specified performance during cycling, it shall be realigned or readjusted once and the test continued. No repair or replacement of parts shall be permitted. If the equipment fails to meet specified performance during testing, or fails subsequently, it does not pass the test. In addition, if the equipment fails to meet specified requirements after final conditioning and adjustment, it does not pass the test.

a. Perform the first four cycles with the equipment within its case.

b. Perform the fifth cycle of this test with the equipment removed from its case. The service selector switch of the RT-834 shall remain in the "Oven On" position. No measurements shall be made until after the 24 hour drying out period.

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4.7.3 Elevation test (see 3.4.3) - The equipment shall be tested in accordance with Method 500, Procedure I, of MIL-STD-810. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.4.3.

4.7.4 Dust test - The equipment shall be tested in accordance with Method 510, Procedure 1, of MIL-STD-810. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.4.4.

a. The equipment shall be operated for the first 3 hours of step 1. During this time, the measurements of 3.4.4 shall be made.

b. After completion of the dust test, the equipment shall meet full specification performance for the measurements specified in 3.4.4. The equipment shall then be opened and there shall be no deterioration of either external or internal parts.

4.7.5 Salt fog test - The equipment (including accessories) in its normal operating position shall be tested in accordance with Method 509, Procedure 1, of MIL-STD-810. The equipment shall not be opened. After the test, the equipment shall meet full specification requirements for the measurements specified in 3.4.5. No rust shall be imposed on the test item by the test facility (see 3.4.5).

4.7.6 Test for equipment vibration - 90 minutes - The radio set in its case, shall be tested as follows to determine conformance to 3.4.6:

a. Secure the equipment on Mounting MT-3140( )/GRC-106 less shock isolators, to a vibration table that can be controlled within 10 percent of the specified amplitude. The equipment shall be mounted in its normal operating position with connection (such as cable assemblies) in place.

b. Vibrate the complete equipment assembly, successively, in three mutually perpendicular directions over a frequency range of 10 to 55 Hz. The total excursion of the applied vibration shall be 0.040 inch in the horizontal plane and 0.060 inch in the vertical plane.

c. In each of three directions, change the frequency in steps of one Hz per second.

d. The equipment shall be subjected to 30 minutes of vibration in each of three mutually perpendicular planes parallel to the equipment surfaces. The tests of 3.4.6 shall be made before and after the test. During vibration, the tests of 3.4.6 shall be performed at two (2) frequencies during each vibration plane for a fifteen (15) minute duration on each frequency. These frequencies shall be selected from the list of frequencies in 4.6.2 and shall be varied between units.

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4.7.6.1 Limited vibration - The Receiver-Transmitter RT-834/GRC in its case, shall be tested as follows to determine conformance with 3.4.6.1:

a. Secure the Receiver-Transmitter RT-834/GRC on Mounting MT-3140( )/GRC-106 to a vibration table that can be controlled to within 10 percent of the specified amplitude. The equipment shall be mounted in its normal operating position, shock mounts installed, with connections (such as cable assemblies) in place.

b. Vibrate the RT unit over a frequency range of 10-55 Hz in the vertical plane. The total excursion of the applied vibration shall be 0.060 inches.

c. The frequency of vibration shall be changed in steps of one Hz per second.

d. The tuned frequency of the RT unit shall be changed at the end of each vibration cycle 10-55 Hz and a different frequency for each successive cycle shall be utilized corresponding to successive frequencies listed under specification paragraph 4.6.2.

e. The RT unit shall be adjusted for CW transmission and the transmission monitored by an oscilloscope placed across the 50 ohm load on the RF drive output of the RT-834/GRC and by an adjacent receiver tuned to the corresponding frequency for the first half of the vibration cycle 10-55 Hz. Noise figure measurement shall be made during the last half of the vibration cycle 55-10 Hz.

f. Measurement of noise figure and CW power output (taken at the RF drive connector across a 50 ohm load) shall be made before and after the test of (a) through (e) above.

g. During the monitoring period of (e) above, there shall be no break-up of the CW transmission during the first half of the vibration cycle 10-55 Hz as observed on the oscilloscope or heard in the receiver. There shall be no noise break-up during the remaining half cycle 55-10 Hz.

4.7.7 Bounce test (see 3.4.7) - The equipment shall be tested in accordance with Method 514, Procedure IX, Part 2, of MIL-STD-810. Prior to testing, the equipment shall meet full specification performance for the measurements specified in 3.4.7. The receiver-transmitter or amplifier, in turn, shall be secured to the 48 inch by 48 inch vehicular adapter plate using Mounting MT-3140 and shock isolators. Before, during, and after each 45 minute phase of step 3, the equipment shall be subjected to the tests of 3.4.7. Rest periods may be provided to permit cooling of the resilient mounts and changing frequency provided that the total testing time is not less than 3 hours.

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4.7.8 Shock test; bench-handling (see 3.4.7) - The chassis-and-front panel assembly shall be removed from its enclosure, as for servicing, and placed in a suitable position for servicing on a solid, 2-inch fir bench top. The test shall be performed as follows, in a manner simulating shocks liable to occur during servicing:

a. Tilt up the assembly through an angle of 30 degrees, using one edge of the assembly as an axis, and permit the assembly to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontal face as an axis, for a total of four drops.

b. Repeat (a) above, with the assembly resting on other faces, until it has been dropped for a total of four times on each face on which the assembly might practically be placed during servicing.

c. At the completion of (a) and (b) above, the equipment shall meet full specification performance for the measurements specified in 3.4.8.

4.7.9 Shock test; ballistic - The test shall be conducted on the Shock Machine for Light Weight Equipment shown in Specification MIL-S-901. The receiver-transmitter or RF amplifier, on Mounting MT-3140( )/GRC-106 shall be secured in its normal operating position to the steel test plate by means of the same fasteners and shock isolators as used for vehicular installation of the equipment. The test shall consist of a total of 9 blows; one each 1-foot blow, 3-foot blow, and 5-foot blow on the back, side, and top of the test plate. As an alternative to re-orienting the test plate for the blows on the side of the plate, equivalent rotation of the equipment under test is permissible. The equipment shall be operated in CW key down condition during each blow. Before and after each blow, the equipment shall meet full specification performance for the measurements specified in 3.4.7.

4.7.10 Acoustic noise test - Using a suitable calibrated sound level meter with a flat (non-weighted) response, acoustic noise shall be checked in all directions at points 2 feet away from the radio set at both low and high blower speeds. The set shall be operating, with standard 2-tone modulated input, and all blowers, air passages, shock mount assemblies, and other devices that have an effect on the acoustic noise level shall be in normal operating condition. The requirements of 3.4.8 shall be met.

4.7.11 Immersion test (see 3.4.9) - The receiver-transmitter or RF amplifier, in turn, shall be immersed to a minimum depth of 3 feet of fresh water for two hours. Immediately prior to immersion, the temperature of the equipment shall be 40°F, or more, above the temperature of the water. The tank in which the equipment is immersed shall be of sufficient capacity to maintain the water

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4.7.11 Immersion test (see 3.4.9) - (contd) - temperature within + 2°F of its initial temperature, or the temperature of the water shall be maintained within those limits by other means. After completion of the 2-hour period of immersion, the equipment shall be removed from the water and wiped dry on exterior surfaces. When the equipment is opened, there shall be no evidence of leakage. Breather valves shall be plugged up during this test.

4.7.12 Tuning knob stability - The torque required to turn the "ANT TUNE" and "ANT LOAD" knobs shall be measured and recorded for clockwise and counter-clockwise rotation. The requirements of 3.4.11 shall be met.

4.7.13 End stops test - A torque of 30 inch-pounds shall be applied so as to check the end stops of the "ANT TUNE" and "ANT LOAD" knobs only. This torque shall be applied 50 times on each end stop direction. The requirements of 3.4.12 shall be met.

4.7.14 Air-seal test - The equipment shall be opened and closed again in such manner as to break and remake the seal. Immediately thereafter, the equipment as field transported shall be subjected to a vacuum of 1 psig (1 pound per square inch less than the atmospheric pressure surrounding the equipment) applied to the interior of the equipment inclosure. The vacuum then shall be valved-off and the interior pressure measured during the ensuing period of 1 minute. The gage used for measurement of the vacuum shall be of such accuracy that a difference of 0.01 pound can be determined readily. The requirement of 3.4.13 shall be met.

4.8 Electrical tests (see 3.6) -

4.8.1 Standard test conditions - Unless otherwise specified, the following standard test conditions shall apply:

4.8.1.1 Atmosphere - All tests shall be made at prevailing factory conditions of temperature, humidity, and pressure, unless otherwise specified in the detailed procedure.

4.8.1.2 27.0-VDC source - The power source shall be 27.0 $\pm$ 0.5 volts, direct current (VDC), except where specifically changed in the detailed procedure. The regulation shall be such as to maintain this voltage when the load is changed from no load to the maximum for any of the equipments under test. Noise and ripple voltage of the source shall be less than 0.5 volts, ac. The voltage source shall be capable of supplying surge currents of 100 amperes for a maximum of 1/2 second without dropping the voltage below 22 volts during the surge. The voltage source shall be adjustable over the range of 22 to 30 volts.

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4.8.1.3 Dummy antenna - A 50-ohm resistive load shall be used except where the detailed procedure requires the use of a whip antenna. The 50-ohm antenna shall be non-reactive.

4.8.1.4 Whip antenna - Where the detailed testing procedure requires a whip antenna, a dummy whip antenna shall be used unless otherwise specified herein. The dummy antenna shall be Dummy Load Group, Electrical OA-4539/GRC-106, which will be loaned to the contractor by the Government.

4.8.1.5 Grounding - A single ground connection shall be provided which shall connect to the internal grounding system of the equipment. Test equipment shall be grounded to the common equipment ground whenever practicable. The cases of metal cased test equipment shall be grounded and so noted in test record.

4.8.1.6 Standard two-tone input - Unless otherwise specified in the detailed test procedure, the modulation applied to the radio set shall be two audio tones equal in amplitude (200 MV (RMS)/600 ohms) and whose frequencies shall be 1500 and 2500 Hz.

4.8.1.7 Standard voice test message - The contractor will furnish a magnetic tape upon which is recorded a test message to be used where specified in the detailed test procedure. The test message will consist of the following phrases repeated over and over; "Jo took father's shoe bench out, she was waiting at my lawn". The test message signal level shall be an average 200 MV (RMS)/600 ohms.

4.8.1.8 Warm-up period - Unless otherwise specified, measurements shall be made after one minute from the time primary power is applied.

4.8.2 Warm-up test - This test shall be started after the set has been in the off condition for at least one hour. With the test set-up for SSB voice power output (4.8.35.4), the service selector switch shall be placed in the standby position for a period of 5 minutes and the "Tune Operate" switch in the tune position. Within 60 seconds after the service selector switch is placed in the SSB/NSK position, the PEP shall be recorded with the "Tune Operate" switch in the "Operate" position; then the service selector switch shall be placed in the Oven On position for 30 minutes. After this, the noise figure (4.8.13) shall be recorded within 60 seconds after the service selector switch is placed in the SSB/NSK position. The test frequency shall be 29.9999 MHz. The requirements of 3.6.1 shall be met.

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4.8.3 Frequency response, transmitter test - The frequency response characteristics shall be determined by applying a tone of constant amplitude (20 MV/600 ohms) to the MT-834 with fixed ALC voltage and varying the frequency from 20 to 6,000 Hz by sufficiently small increments to give a smooth curve of the audio response of the overall equipment. The measuring device may be the indicator of rf power into the dummy load or the spectrum analyzer as used in the intermodulation distortion (4.8.4) tests. This frequency response test shall be conducted at 2 mc. The frequency response of this equipment shall conform to 3.6.2.

4.8.4 Intermodulation distortion test - The intermodulation distortion shall be measured for conformance to 3.6.3. Measurements shall be made at frequencies to cover all band edges and centers.

4.8.5 Hum and noise test - This test shall be made at the same time as the intermodulation distortion test (4.8.4). The two audio tones shall be as specified in 4.8.1.6. The requirements of 3.6.4 shall be met.

4.8.6 Phase modulation and jitter test - This test shall be performed during the bounce test (4.7.8) in both the transmit and receive modes. In the receive mode, the audio distortion shall be measured. The results shall comply with the requirements of 3.6.31. In the transmit mode, the output shall be observed on a spectrum analyzer and oscilloscope. The results shall comply with the requirements of 3.6.5.

4.8.7 Carrier suppression test - This shall be measured at all frequencies listed in 4.6.2, using a spectrum analyzer connected to the output of the radio set through suitable attenuators. The two audio tones shall be as specified in 4.8.1.6. The requirements of 3.6.6 shall be met.

4.8.8 Opposite sideband suppression test - This shall be measured at all frequencies listed in 4.6.2 during the test of 4.8.7 above. The requirements of 3.6.7 shall be met.

4.8.9 CW keying tests (see 3.6.8) - Keying shall be at 25 dot cycles per second (60 wpm).

a. Reception: Distortion of the received CW signal shall be measured.

b. Transmission: The rf output wave shape shall be checked for conformance to the requirements of 3.6.8(b) and (c). Check number of Hz that the CW frequency is above the carrier frequency.

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4.8.10 Signal level meter test - With the AGC switch in the ON position and the set alternately keyed and unkeyed, the reading on the meter shall be recorded as the input to the set is increased in 20 db steps from 5 uv to 0.5 volts. The requirements of 3.6.9 shall be met.

4.8.11 BFO test (see 3.6.10) - This test shall be made during the temperature test of 4.7.1. The BFO control shall be varied from full CCW to full CW and the resultant audio frequency recorded for each extreme. The requirements of 3.6.10 shall be met.

4.8.12 Overall gain test - With the set in the SSB/NSK mode, a signal of four microvolts shall be fed into the antenna input using a 75-ohm source. The signal shall be an unmodulated carrier adjusted in frequency to give a 1,000 cycle beat note with the local carrier. The output shall be measured at all the frequencies listed in 4.6.2. The requirements of 3.6.11 shall be met.

4.8.13 Noise figure test (see 3.6.12) - The noise figure of the receiver shall be measured with a noise generator having an output impedance of 75 ohms (Rhode Schwarz Noise Generator Type SKTV Code BN4151/75, or equal). The noise figure shall be measured using the test frequencies of 4.6.2. The measurements shall be made at the IF output of the RT-834. The audio output shall be monitored during this test.

4.8.14 Frequency response, receiver test - The service selector switch shall be set in the SSB/NSK position. An unmodulated signal shall be applied to the antenna input terminal from a 75-ohm source. With the AGC switch in the OFF position, the signal generator shall be tuned to produce a signal in the IF amplifier. With a suitable signal input level, the gain control shall be set to produce a reference output of 10 mw at the receiver's audio output. The signal input shall be increased by 3, 6, 10, 20, 40, 60, and 80 db. At each step, the generator shall be detuned to each side of resonance until the reference output is again observed. Frequency deviation from resonance for each step shall be measured and recorded and a selectivity curve drawn. The test frequency shall be 2.0000 MHz. The requirements of 3.6.2 shall be met.

4.8.15 Tuned circuit mistracking - The tracking error of the RF amplifier module shall be measured as follows:

a. Remove the translator module and connect VTVM with 50Ω load to connector A8-J1B-A2.

b. Connect signal generator to receiver input through a variable attenuator and 25Ω series resistor.

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4.8.15 Tuned circuit mistracking - (contd) -

c. Tune the RT-834 to one of the six (6) test frequencies.

d. Tune signal generator for exactly the same frequency and check with counter. Disconnect counter after checking frequency and adjust level for 1 mv at the receiver antenna connector.

e. Measure and record VTVM reading in db above 1 mv (R.F. Amplifier Gain).

f. Vary the signal generator around the starting frequency to obtain a peak reading on the VTVM.

g. Measure and record, in db, the increase in the VTVM reading.

h. Measure and record the frequency difference and sign (plus or minus) at which the peak reading was obtained.

i. Repeat steps c through h above for the remaining five (5) frequencies.

4.8.16 Desensitization test - During the test for cross modulation (4.8.27), the undesired signal separation that reduces the desired audio output reference level by 3 db shall be measured and recorded. The requirements of 3.6.14 shall be met.

4.8.17 IF bandwidth test - Using a stable rf signal generator covering the IF frequency, a counter, and a calibrated vacuum-tube voltmeter (VTVM), the IF bandwidth characteristics shall be measured as follows:

a. With the AGC switch "OFF" and the translator module removed, a 1.752 MHz signal shall be injected, through variable attenuators, into A8-J1A-A3 connector on the chassis and the level adjusted for a 10 mv (0 db reference) at the IF output connector. This level shall be maintained at the output throughout the test.

b. Slowly vary the frequency of the Signal Generator from 1.750300 MC to 1.753500 MC and record maximum and minimum IF output with respect to the 0 db reference level.

c. Re-establish reference level at 1.752 MC, if necessary, before proceeding with test.

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4.8.17 IF bandwidth test - (contd) -

d. Decrease attenuation of Variable Attenuators by 2 db and determine the two frequencies at which the IF output drops to the reference level.

e. Repeat Step d for 3, 6, 20, 30, 40, 50 and 60 db reductions in attenuation.

f. Plot data obtained to show IF frequency response and compare with IF Bandwidth specification.

4.8.18 IF output test (see 3.6.16) - The output shall be obtained after the IF filter. The measurements shall be made with an rf voltmeter. The equipment shall be tuned with a signal input of 4 uv at the asterisked frequencies listed in 4.6.2 and the IF output voltage measured and recorded.

4.8.19 Squelch test - With the receiver-transmitter tuned to 2.0000 MHz (fo) in the receive mode and squelch ON, the receiver must unsquelch when a signal of 0.3 microvolts at fo + 500 Hz is applied to the receiver input. When the applied RF signal is reduced by 20 db, the receiver shall squelch in less than 5 seconds.

4.8.20 AGC and blocking - The audio gain shall be set to produce 1 MW power output with a signal input of 5 uvolts. Change in power output in db shall be recorded as the signal input is increased in steps to 1000 uvolts, and to 1.0 volt. The AGC hang time shall be measured by suddently reducing the signal level to 5 uvolts and noting the time required for the output to return to 1 MW. The AGC attack time shall be measured by suddenly increasing to 1.0 volt and noting the time duration of the overshoot. Test frequencies shall be 2.0, 3.0, 6.0, 12.0, 24.0 and 29.9999 MHz. The requirements of 3.6.18 shall be met.

4.8.21 Frequency standard adjustment and calibration tests - A sample of the frequency standard output shall be taken at the connector on the front panel labeled "Freq. Std.". A test shall be performed, using a suitable counter and VTVM attached to this connector with Radio Frequency Cable RG-58( )/U. The requirements of 3.6.19 shall be met.

4.8.22 Internal and external standards tests - The level of the 5 MHz Frequency Standard output, terminated with 50 ohms, shall be measured using a suitable VTVM. The frequency accuracy test (4.8.26) shall be performed with the radio set locked to an external standard. The results of the tests shall comply with the requirements of 3.6.20.

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4.8.23 Frequency stability test (see 3.7.21) -

4.8.23.1 Long term stability test - A long term stability test shall be made. The accuracy of the test equipment shall be 1 part in 10^8 per week. A sample of the 5 MHz Frequency Standard output shall be measured and recorded every 24 hours. The requirements of 3.6.21.1 shall be met.

4.8.23.2 Short term stability test - A short term stability test shall be made. The accuracy of the test equipment shall be 5 parts in 10^8 per hour. The test shall be made at 5 MHz. The requirements of 3.6.21.2 shall be met.

4.8.23.3 Warm-up from hot oven test - The oven shall be turned on for one hour. The test of 4.8.23.2 shall be made at 10 MHz. The requirements of 3.6.21.3 shall be met.

4.8.23.4 Warm-up from cold start - During the low temperature cycle test 4.7.1.1, the 5 MHz Frequency Standard shall be measured using test equipment with an accuracy of 1 part in 10^7 per hour. The requirements of 3.6.21.4 shall be met.

4.8.23.5 Overall frequency stability test - The test of 4.8.23.2 shall be repeated with the radio set unlocked. The overall frequency drift, the proportional drift during one hour, and any sudden changes in frequency shall be noted. The requirements of 3.6.21.5 shall be met.

4.8.23.6 Mobile stability test - The test of 4.8.23.2 shall be made during the equipment vibration test (4.7.7) and the bounce test (4.7.8). In addition, the frequency setting and the output frequency shall be noted before, during, and after the test. The requirements of 3.6.21.6 shall be met.

4.8.23.7 Stability vs line voltage test - After stabilization at 15 MHz with 27.0 vdc input, the input shall be changed to 22.0 volts and the frequency checked immediately and each 5 minutes for 1/2 hour. The input shall then be switched to 30.0 vdc, and the frequency checked in a similar manner. The requirements of 3.6.21.7 shall be met.

4.8.24 Frequency accuracy test - The actual transmitting carrier frequency shall be measured at all the frequencies listed in 4.6.2. The requirements of 3.6.22 shall be met.

4.8.25 Vernier tuning test - A check shall be made at 4.9986 MHz using the 5 MHz standard as the received signal source to determine compliance with 3.6.23.

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4.8.26 Internal signals test - The radio set shall be tuned over its frequency range in the SSB mode with the AGC off. All internal signals that produce an increase of the audio output, 10 db or more above the receiver noise level, shall be noted. The audio signal level shall be noted and the radio set then detuned approximately 10 KHz or to a frequency that does not produce an internal signal. A signal generator shall then be tuned to the new frequency and the generator output level adjusted to produce the same audio signal level. The signal generator output in microvolts shall be recorded as the equivalent strength of the internal signal. The radio set shall be checked at every 100 Hz detent from 2.0000 to 29.9999 MHz. The requirements of 3.6.24 shall be met.

4.8.27 Cross-modulation test - The service selector switch shall be set in the SSB-NSK position. The AGC switch shall be set in the ON position. Desired and undesired signals shall be fed into the receiver-transmitter under test from separate 75-ohm signal sources. The generator furnishing the desired signal shall be first connected to the receiver-transmitter set to 100 uv, and tuned to the test frequency (fo) to produce an IF output reference frequency of 1.752 MHz and an audio reference frequency of 2 KHz. The receiver "S" meter indication shall be noted. The desired signal shall then be connected to the receiver-transmitter through a 10,000-ohm resistor and the generator signal raised to again produce the reference IF, audio, and "S" meter levels. The generator furnishing the undesired signal shall be connected directly to the receiver-transmitter and its output adjusted for 30 percent, 1 KHz modulation at levels of 0.01, 0.1, 1.0, and 6.0 volts. All undesired signal levels shall be open-circuit measurements. At each level, the undesired signal shall be tuned toward the desired signal, from each side, until 10 percent modulation is produced on the desired IF signal. (Undesired signals 26 db below desired signals). A selective voltmeter with a modulation meter or a spectrum analyzer may be used for this measurement at the IF output. The frequency separation between the desired and undesired signals, for each such set of conditions, shall be measured and recorded. The reference audio level shall be observed. If a lower reading is obtained, desensitization has occurred and should be noted as such. The results shall be within the limits of 3.6.25. Not more than 6 test frequencies, designated by the Contracting Officer's Technical Representative, shall be utilized for this test.

4.8.28 IF rejection test - The service selector switch shall be set in the SSB position. The AGC switch shall be set in the OFF position. At the test frequency, the signal generator and equipment shall be adjusted so that 1.0 milliwatt audio output is obtained from the radio set with a signal input of 0.5 microvolts. The signal generator shall then be tuned to the interference frequency (fixed and variable IF frequencies) and, without altering the equipment controls, the signal generator output shall be increased until 1.0 milliwatt power output is again produced. The ratio between the initial and final signal generator output in db shall be recorded as the rejection ratio. The requirements of 3.6.26 shall be met.

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4.8.29 Image rejection test - The test procedure shall be the same as that used for IF rejection (4.8.28). The requirements of 3.6.27 shall be met.

4.8.30 Crossover rejection test - The test procedure shall be the same as that used for IF rejection (4.8.28). The requirements of 3.6.28 shall be met.

4.8.31 Receiver-SSB intermodulation test - This test shall be performed with signal inputs of 5.0, 100, 1,000, 10,000 and 100,000 uv for each tone. Measurements shall be made in accordance with 3.6.29.

4.8.32 BFO and carrier leakage test - With a signal input at the receiver of 4 uv, bfo and carrier leakage shall be measured across a 50-ohm (nominal) impedance at the IF output connection. The requirements of 3.6.30 shall be met.

4.8.33 Audio distortion - With a receiver input signal of 1, 1000, and 1,000,000 microvolts at a frequency of fo + 1000 Hz, the audio distortion shall be measured at the 2.0 watt and 10 mw output levels.

4.8.34 Audio power test - During the test of 4.8.12, the front panel control shall be operated from its minimum to the maximum positions and the results noted. The requirements of 3.6.32 shall be met.

4.8.35 RF power output tests (see 3.6.33) - The rf power output shall be measured, unless otherwise specified, using the standard conditions of 4.8.1. All rf power output measurements shall be made as functions of the voltage across the resistive element of the dummy antenna, and shall be corrected for temperature and frequency characteristics. Correction curves for frequency effects shall be furnished by the contractor for all rf meters used on these tests.

4.8.35.1 Power vs frequency test - The power shall be measured over the entire frequency range into the dummy antenna (4.8.1.3). The equipment shall be modulated with the standard two-tone input (4.8.1.6). The equipment shall be retuned at frequencies to cover all band edges and centers and the power output measured at each point. The frequency of lowest power output shall be specifically noted. The complete test shall be repeated for CW (4.8.35.6) and AM (4.8.35.5) modes of emission. The power output shall conform to the requirements of 3.6.33.1.

4.8.35.2 Antenna tuner test - The power output shall be measured at the frequencies specified in 3.6.33.2. The equipment shall be modulated with the standard two tone input (4.8.1.6) and the dummy whip (4.8.1.4) shall be used. The tuner test shall be repeated using the asterisked frequencies listed in 4.6.2, into a doublet antenna having a 2.5:1 VSWR. The requirements of 3.6.33.2 shall be met.

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4.8.35.3 Power vs temperature test - During the temperature cycle test (4.7.1), the RF power shall be measured at the various temperatures in accordance with steps in (4.7.1). The dummy whip antenna (4.8.1.4) and the frequency which gave the lowest power output during the test of 4.8.35.2 shall be used. All other conditions shall be standard (4.8.1). The output shall be measured using the two audio tones of 4.8.1.6 in the SSB mode. The test shall be repeated at no more than 7 additional frequencies specified by the Contracting Officer's Designated Representative from the list of frequencies in 3.6.33.2. The requirements of 3.6.33.2 shall be met except that at temperatures above 55°C, the power output measured may be 0.5 db down from the power noted for the test frequencies selected for this test in 3.6.33.2 (see 3.6.33.3).

4.8.35.4 SSB power output test - Using the standard two-tone input (4.8.1.6), power output into a dummy antenna (4.8.1.3) shall be noted at all frequencies listed in (4.6.2). The test shall be repeated using the standard test message (4.8.1.7). The requirements of 3.6.33.4 shall be met.

4.8.35.5 Compatible AM test - The PEP and the carrier frequency power shall be measured using the standard test message (4.8.1.7), at the frequencies listed in 4.6.2 into a dummy antenna (4.8.1.3). The requirements of 3.6.33.5 shall be met.

4.8.35.6 CW and FSK test - In both the CW and the FSK modes, the average power output shall be measured into a dummy antenna (4.8.1.3) at the frequencies listed in 4.6.2. During the FSK test, a 200 MV, 2 KHz tone shall be applied to the 600 ohm input. The requirements of 3.6.33.6 shall be met.

4.8.35.7 Power vs line voltage test - The SSB test (4.8.35.4) on two-tone only shall be repeated using inputs of 22.0, 24.0, 25.0, and 30.0 volts dc. The requirements of 3.6.33.7 shall be met.

4.8.36 Interlock test - The Amplifier AM-3349 shall be pulled from its case, 1/2 inch and the interlock operation noted. After the unit is replaced in its case, the equipment shall be checked for proper operation (see 3.6.34).

4.8.37 P.A. overload test - The PA overload circuit shall be checked by applying overloads of 25 and 200 percent, then the overload reset shall be checked. The requirements of 3.6.35 shall be met.

4.8.38 Reverse current test - With the radio set turned ON, the input potential shall be reversed 6 times with a one minute on for each reversal. All evidence of damage or abnormal operation shall be noted. After the reversals, the noise figure test (4.8.13) and the SSB power output (4.8.35.4) check shall be performed (see 3.6.36). This test shall be performed at 2.0000 MHz only.

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4.8.39 Oven protection test (see 3.6.37) - Proper operation of the oven protection circuit shall be checked as follows:

a. Remove the crystal oven from the frequency standard module, place in the temperature chamber and attach ohmmeter leads to pins J6 and J7. Note that there is continuity at room temperature.

b. Adjust chamber temperature to 95°C and allow a 1/2 hour stabilization period.

c. Verify with ohmmeter that the thermal switch is now open (open circuit between pins J6 and J7).

4.8.40 Antenna transfer test - All antenna transfer switching shall be checked from full rf output to receive and in the reverse direction. Safety devices on output connectors shall be checked at this time. The requirements of 3.6.38 shall be met.

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4.8.41 Power drain test. - With the transmitter in any mode of operation, the total current drain shall be noted at 27.0 volts, dc input, for all frequencies listed in 3.6.33.2. The requirements of 3.6.39 shall be met.

4.8.42 VOX operation test (See 3.6.40). - The service selector switch shall be set in the SSB/NSK position. With the VOX switch in the push-to-talk position, the radio set shall switch to transmit when the keyline is grounded. When the VOX switch is in the push-to-vox position and a 10 mv, 500 Hz signal is applied to pins J and D of the 10-pin audio connector, the radio set must switch to transmit when the keyline is grounded. When the signal is reduced to 5 mv, the radio set shall switch to receive and grounding the keyline shall not switch the radio set to transmit. When the VOX switch is in the VOX position, the radio set shall not switch to transmit with the 5 mv signal but shall switch to transmit when the signal is raised to 10 mv.

4.8.43 Metering tests. - All meters shall be checked for proper readings in all positions of the appropriate meter switch. These tests may be made concurrently with the appropriate electrical test. (See 3.6.41).

4.8.44 TUNE power output test. - The average power output in the TUNE position shall be measured at the frequencies listed in 4.6.2 into a dummy antenna (4.8.1.3) with no modulation applied to the input of the radio set. The requirements of 3.6.42 shall be met.

4.8.45 CW Sidetone test (See 3.6.43). - With the radio set in CW transmit mode, a CW sidetone shall be present at the audio output. The level shall be adjustable by the audio gain control. The CW sidetone frequency shall be adjustable by the BFO control and the frequency limits shall be the same as those for the BFO control (3.6.10).

4.8.46 Receiver front end protection. - A transmitter or suitable substitute shall be connected to the AN/GRC-106 whip antenna connector through a suitable attenuator to simulate the effects of a high power transmitter operating in proximity to the radio set. The radio set and above transmitter shall be set to 2.0000 MHz. The rf input voltage to the AN/GRC-106 in the receive mode, shall be increased gradually from 0 to 9 volts as measured at the input of the AN/GRC-106. Operation of the protection circuit will be noted as a sharp change in input voltage from 7 volts to 16 volts. The requirements of 3.6.44 shall be met.

4.9 Inspection for dimensional interchangeability. - Radio sets which have been subjected to and passed all Group A inspections and Group B inspections shall be tested for compliance with the requirements of paragraph 3.3.5. The specified number of sample units shall be selected from the lot and the modules therein shall be identified with the serial number of the unit from which they were removed. The mechanical dimensions required by the gage drawings shall be measured in accordance with gage list SC-GL-323149 to

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determine compliance with the mechanical interchangeability requirements. Modules which fail to meet the specified gage limits shall be considered not acceptable. Radio sets which fail to meet the dimensional interchangeability requirement or which contain modules which fail to meet the specified gage limits shall be considered to have failed the mechanical interchangeability requirement.

4.10 Electrical Interchangeability. - Radio sets which have been subjected to and passed Group A and Group B inspections shall be tested to determine compliance with the requirements of paragraph 3.3.5. The plug-in modules and sub-assemblies listed below shall be tested to determine compliance with the performance requirements specified in the applicable test procedure of the Government owned electrical test fixtures and requirements of the corresponding USAECOM test specifications (See 3.3.5).

RT-834 Modules:

1A1	Panel and Chassis
1A1A1	100 Hz synthesizer
1A2	100 KHz synthesizer
1A3	Frequency standard
1A4	10 KHz and 1 KHz synthesizer
1A5	Transmit IF and audio
1A6	Frequency dividers
1A7	Receive IF
1A8	Translator
1A9	MHz synthesizer
1A10	Receiver audio
1A11	Dc to dc converter
1A12	RF amplifier

AM-3349 Amplifier Modules

2A1	Chassis and plenum
2A2	Turret
2A3	Antenna coupler
2A4	Discriminator
2A5	Front panel
2A6	Case blower and converter
2A7	Relays
2A8	Driver
2A9	PA stator

4.11 Fungus test. - The equipment (including accessories) shall be tested in accordance with Method 508, Procedure 1, of MIL-STD-810. Spraying shall include parts and materials on and under the chassis, and the interior of the equipment case. The equipment shall remain in an open condition during the test. Operation is not required (3.3.4).

4.12 Visual and mechanical inspection. - Visual and mechanical inspection shall be in accordance with MIL-STD-252 and the following inspection: parts, mechanical construction, loose nuts and screws, application of specified treatments and miscellaneous defects. The equipment and constrols shall be inspected for proper mechanical and electrical operation. Wiring, soldered connections, grounds, welds, finishes, etc. shall be inspected for workmanship. Clearances, dimensions, and mechanical adjustments shall be measured. Labels and markings shall be inspected for conformance with approved facsimiles, when facsimiles are required. Examination also may be made for other visual or mechanical defects, similar to those described above, that are contrary to specified requirements for the equipment. Visual and mechanical defects shall

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be classified by the Government as major or minor in accordance with the definitions of MIL-STD-109.

4.13 Operational inspection (See 3.7). - A standard radio set (AN/GRC-106 or equivalent) shall be used for talking with (by radio) to a set under test. The bellows connector and connecting cables to be shipped with the end item shall be used during this test. The set under test shall be checked for the following:

a. Proper meter reading when switch is alternately set for each mode of operation.

b. Porper mechanical insertion and electrical operation of Bellows Connector (the one to be shipped with the radio set).

c. Proper electrical transfer and mechanical function of antenna flag switch.

d. Proper electrical function of BFO control.

e. Proper operation of heat exchanger (Note: on one (1) out of fifteen (15) sets tested, the air flow shall be restricted to create additional heat which should energize the high speed operation of the fan and motor assembly.

f. Proper operation of the vacuum relay-electrical transfer.

g. Proper electrical operation of the frequency vernier, AM, squelch, vox and push to box controls.

h. Proper operation of CW on any one of the frequencies listed in 4.6.2.

i. Proper transmission and reception of SSB voice (by radio) shall be ascertained on the frequencies listed in 4.6.2 (Note: When alternately adjusting the Antenna Tune and Antenna Load controls, check antenna meters for any sudden change in direction of meter needle which indicates a defective or intermittent roller contact on Coil L-1).

j. Momentary HV protect trip-outs shall be noted during the entire operational inspection. Trip-outs of the protect circuit when no circuit failures can be found constitutes a failure.

4.14 Quality conformance inspection of preparation for delivery. - Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of section 5.

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4.15 Rough handling test (preparation for delivery). - When a rough handling test, in accordance with Specification MIL-P-116 is required by the contract (See 6.2 (d)), the following functional tests shall be conducted to determine freedom from operational malfunction caused by rough handling:

Noise figure (4.8.13)
SSB power output (4.8.35.4)

4.16 Reliability. - The sample units shall be tested in accordance with Appendix A to this specification and shall meet the requirements of 3.10.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. - Preservation and packaging shall be level A or C as specified (See 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. - Radio Set AN/GRC-106( ) (units of) shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. - Radio Set AN/GRC-106( ) (units of) shall be dried in accordance with the applicable procedures of MIL-P-116.

5.1.1.3 Preservation application. - None required.

5.1.14 Unit packaging. - Unit packaging shall be in accordance with methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Spare fuses. - Spare fuses shall be packaged method III as follows: Cushion the fuses by wrapping together in paperboard conforming to PPP-P-291, type I. Secure wrap with gummed paper tape conforming to PPP-T-45, type III, grade A. Secure packaged fuses to handle of receiver-transmitter with tape conforming to PPP-T-76.

5.1.1.4.2 Receiver-Transmitter, RT-834( )/GRC. - Each item shall be packaged individually in accordance with items 1 through 6 of Table VIII and figure 3.

5.1.1.4.3 Amplifier, Radio Frequency, AM-3349( )/GRC-106. - Each item shall be packaged individually in accordance with items 1 through 7 of Table IX and figure 4.

5.1.2 Level C. - Radio Set AN/GRC-106( ) (units of) shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration and damage during shipment, handling and limited intransit storage.

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5.2 Packing. - Packing shall be level A, B or C as specified. Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A.

5.2.1.1 Palletized load. - A quantity of Radio Set AN/GRC-106( ) (units of) packaged as specified in 5.1, bearing the same stock number shall be placed on a pallet, load type 1, conforming to MIL-STD-147 except that the pallet shall be softwood conforming to NN-P-71, type IV, size 2. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping.

5.2.1.2 Less than palletized load. - When quantities per destination are less than a pallet load, the item packaged as specified in 5.1 bearing the same stock number, shall be waterproofed with tape conforming to PPP-T-76, in accordance with the taping requirements of the appendix of the box specification. A quantity of the waterproofed containers shall be packed within a close-fitting box conforming to PPP-B-601, overseas type: PPP-B-621, style 4, class 2: or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type 1, class B.

5.2.2 Level B.

5.2.2.1 Palletized load. - A quantity of Radio Set AN/GRC-106( ) (units of), packaged as specified in 5.1, bearing the same stock number, shall be palletized as specified in 5.2.1.1.

5.2.2.2 Less than palletized load. - When quantities per destination are less than a pallet load, a quantity of Radio Set AN/GRC-106( ) (units of), packaged as specified in 5.1, bearing the same stock number, shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistant, variety DW. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, type IV. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or nonmetallic

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strapping conforming to PPP-T-97, type IV and PPP-S-760, type II, respectively: selection of the material and application shall be in accordance with the appendix of the applicable box specification.

5.2.3 Level C.

5.2.3.1 Palletized load. - A quantity of Radio Sets AN/GRC-106( ) (Units of), packaged as specified in 5.1, bearing the same stock number, shall be palletized as specified in 5.2.1.1 except that the fiberboard caps shall be class domestic.

5.2.3.2 Less than palletized load. - When quantities perdestination are less than a pallet load, a quantity of Radio Sets AN/GRC-106( ) (units of), packaged as specified in 5.1, bearing the same stock number, shall be packed as specifi- in 5.2.2.2 except that the fiberboard boxes shall conform to PPP-B-640 and PPP-B-636, class 1 and class domestic, respectively, and reinforcing shall not be required for boxes conforming to PPP-B-636.

5.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. - Radio Set AN/GRC-106( ) is intended to provide vehicular radio communication between similar sets. It will usually be mounted in a 1/4 ton, 4 x 4 truck or a tracked vehicle. A total of 280,000 channels is provided for.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Levels of preservation, packaging, and packing required. (See section 5).

c. Type required.

d. If rough handling test and functional test are required (See 4.15).

e. First Article Inspection:

(1) Six sample units of each item cited in section 1 are generally required so that lengthy environmental tests can be completed on three sample units while complete performance measurements can be made on the other sample units. (See 3.2).

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(2) Detailed requirements covering First Article packs, including inspection to be performed thereon and descriptive details, plans, drawings, etc. (NOTE: These may be obtained from Packaging Branch, Materiel Management Directorate, US Army Electronics Command, Philadelphia, Pennsylvania 19103).

f. Marking shipping of samples.

g. Place of final inspection.

h. Technical literature required. (See 3.8).

6.3 Color (See 3.4.1). - The color chip furnished by the procuring agency will be color chip No. X-24087, and may be obtained upon request to Commander, US Army Electronics Command, 225 South Eighteenth Street, Philadelphia, Pennsylvania 19103.

6.4 Humidity. - Natural climatic extremes may include 100-percent relative humidity, with condensation, up to 4 hours at a time. This requirement (See 3.4.2) is specified by Military Standard MIL-STD-210. The moisture-resistance test (see 4.7.2) using Military Standard MIL-STD-810, is a procedure that gives a practical test approximating the natural conditions.

6.5 Definitions of Level B. - Level B packaging requirements detailed herein are designed to furnish protection against corrosion, deterioration, and physical damage during domestic shipments; covered storage for a period of three years or less; and distribution within the continental limits of the United States. (See 5.2.2).

6.6 Group C inspection. - Approval to ship may be withheld, at the discretion of the Government. (See 4.5.3.2).

6.6.1 Initial shipments. - Initial shipments of end items will be contingent only upon successful completion of all Group C tests. (See 4.5.3).

6.7 Location of air seal and operational inspection. - It is desirable that the air seal test (4.7.15) and the operational test (4.13) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is completed. Any preparation for shipment which would require breaking of the equipment seal should be accomplished prior to the air-seal test so that the seal may remain intact thereafter.

6.8 Nomenclature. - The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: AN/GRC-106W. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1).

6.9 Color of textiles (See 3.4.1). - Standard samples of Olive Drab No. 7

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may be obtained from the Sample Loan Unit, Military Clothing and Textile Supply, Philadelphia, Pennsylvania. When requesting samples, indication should be given as to type of material concerned.

6.10 Verification inspection (See 4.5). - Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.11 Dimensional data. - Dimensions of packaging material in the Bill of Materials Tables are based on the dimensions of the equipment cited in the tables. When the dimensions of the item vary, the dimensions of the packaging material shall be adjusted accordingly.

6.12 Deviations. - Request for deviations in packaging and packing design shall be supported by test data proving the ability of the equipment, packaged and packed in accordance with the proposed design change, to withstand the applicable tests for the selected method, including the rough handling tests in accordance with MIL-P-116. Deviation approval shall be contingent on availability of packaging and packing materials selected and economies realized (see section 5).

Custodian:
Army-EL

Preparing Activity
Army-EL

Project Number 5820-A597

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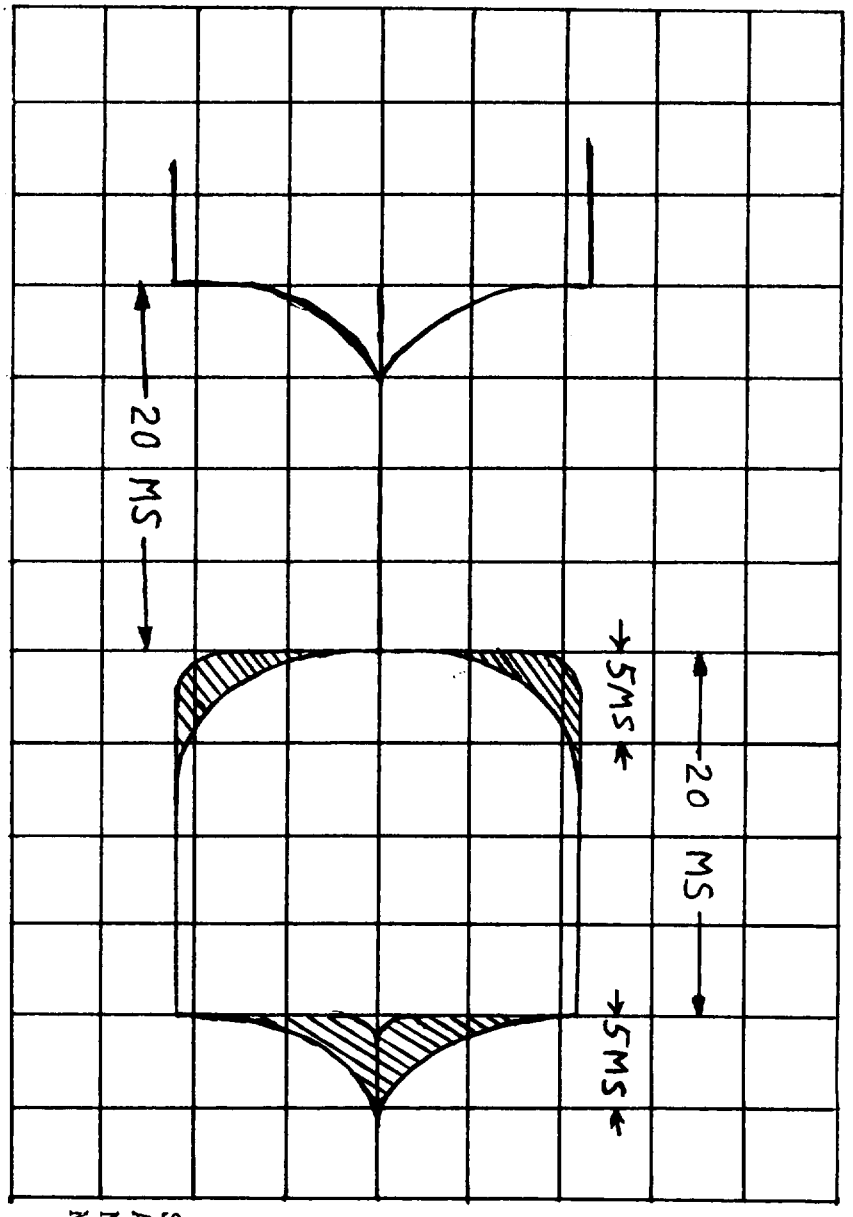


FIGURE I IF OUTPUT CW PULSE SHAPE AT 25 DOT CYCLES

SHADED AREA DENOTES
ACCEPTABLE LIMITS
NO OVERSHOOT PERMITTED
MS=MILLISECONDS

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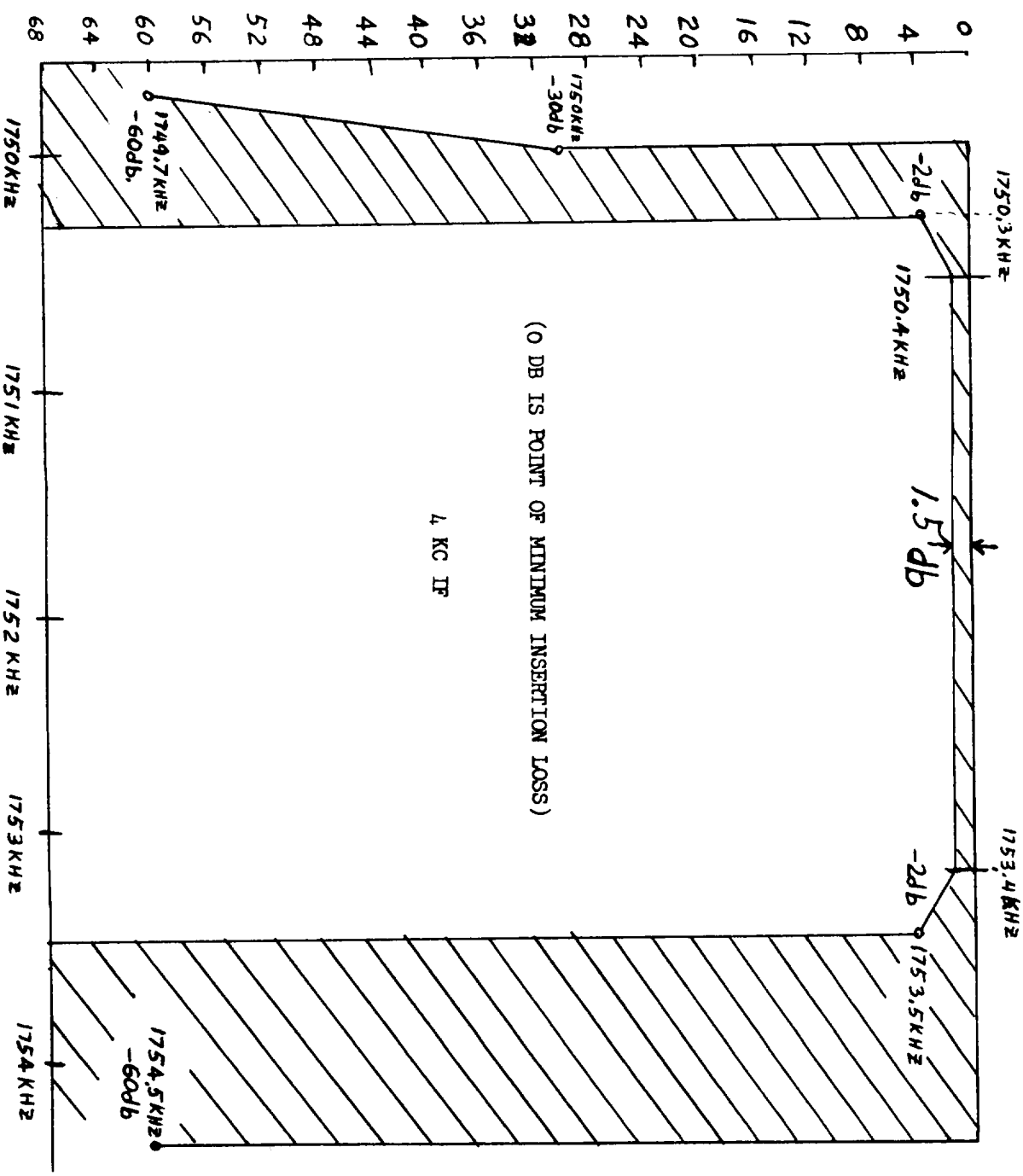


FIGURE 2

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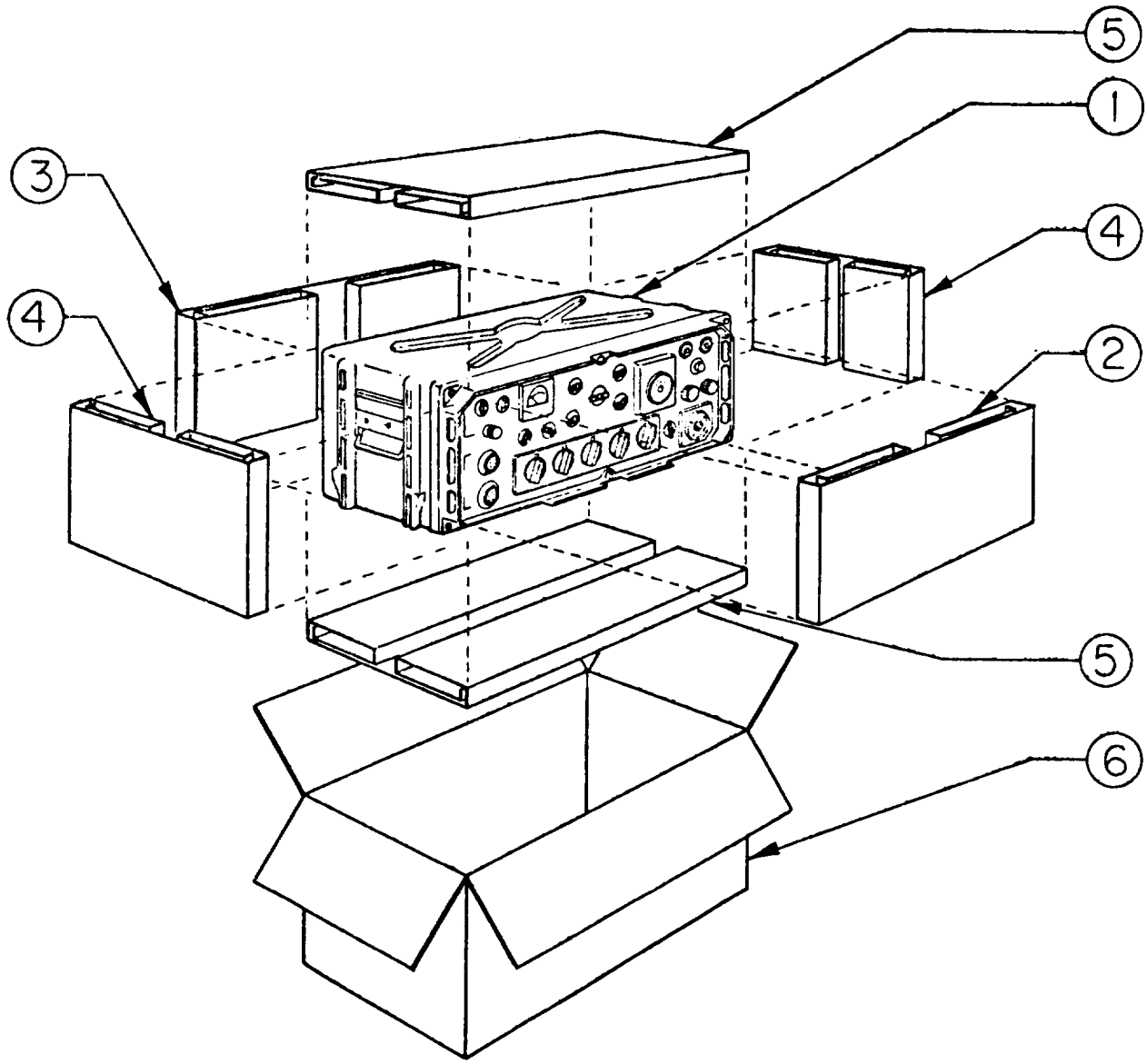


FIGURE 3

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BILL OF MATERIAL
TABLE VIII

REF No.	ITEM Nomenclature	No. Reqd.	Method MIL-P-116	SIZE - INCHES L	W	D	REQUIREMENT Spec No.	T	C	V	G	F	S
1	Receiver-Transmitter, Radio RT-834( )/GRC	1	111	18	13	7	(Bare wt 46 lbs)						
2	Cell, Front	1		14	6-1/2	1-1/2	PPP-F-320	CF	WR	SW	W5c		
3	Cell, Back	1		18	7	3/4	PPP-F-320	CF	WR	SW	W5c		
4	Cell, End	2		15-1/4	7	15/16	PPP-F-320	CF	WR	SW	W5c		
5	Cell, Top and Bottom	2		19-7/8	15-1/4	1-5/16	PPP-F-320	CF	WR	SW	W5c		
6	Box, Fiberboard	1		19-7/8	15-1/4	9-5/8	PPP-B-636	CF	WR	SW	V3c		

Lev. A - PACKED WEIGHT
Lev. A - PACKED VOLUME

T - Type
C - Class
V - Variety
G - Grade
F - Flute
S - Style

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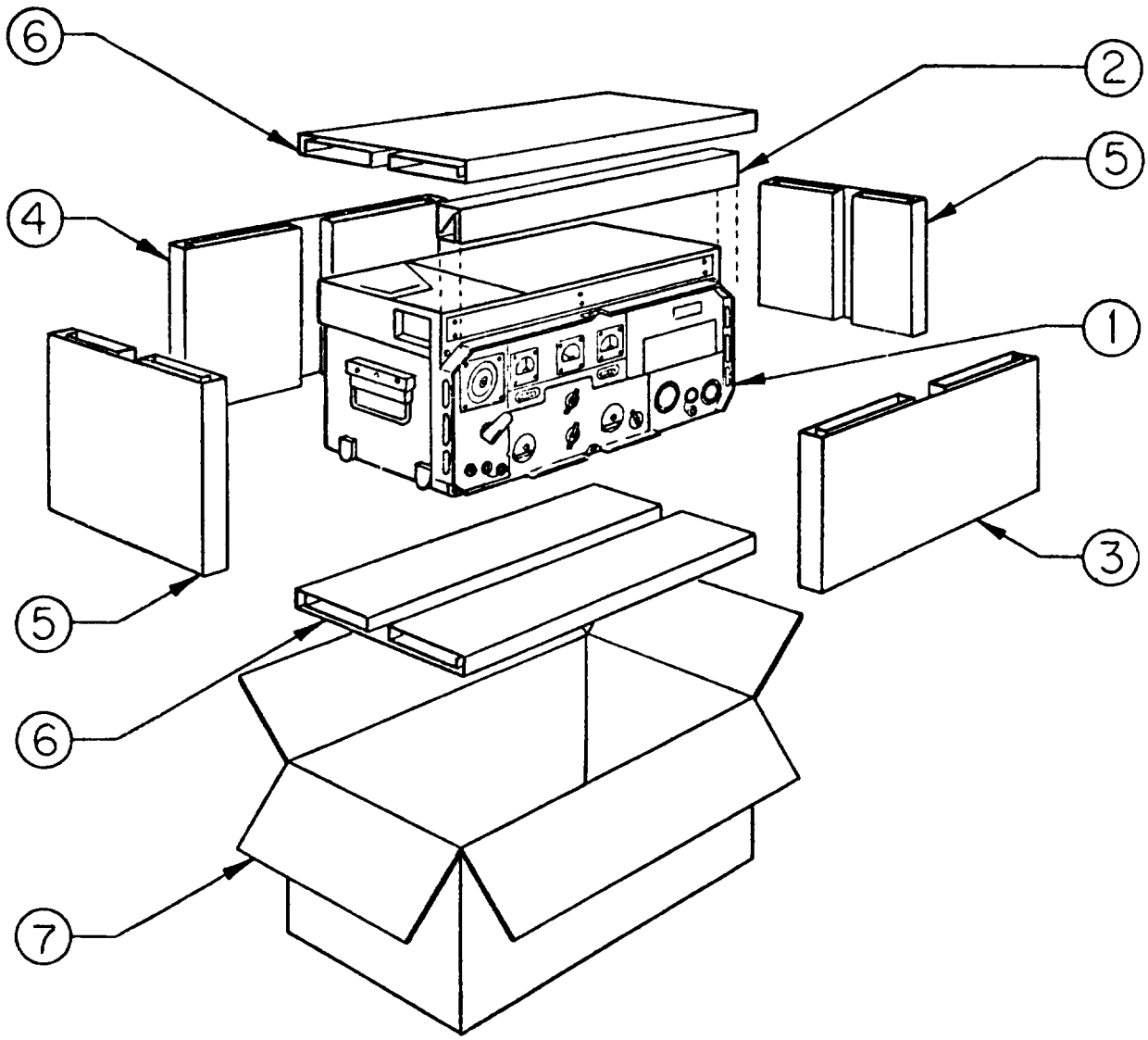


FIGURE 4

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BILL OF MATERIAL
TABLE IX

REF No.	ITEM Nomenclature	No. Reqd.	Method MIL-P-116	SIZE - INCHES L	W	D	REQUIREMENT Spec No.	T	C	V	G	F	S
1	Amplifier, Radio Frequency AM-3349()/GRC-106	1	111	18	13	9	(Bare wt 68 lbs)						
2	Sleeve, Filler	1		18	2	2	PPP-F-320	CF	WR	SW	V3c		
				Place in logging chart cutout.									
3	Cell, Front	1		14	6-1/2	1-1/2	PPP-F-320	CF	WR	SW	V3c		
4	Cell, Back	1		18	9	1	PPP-F-320	CF	WR	SW	V3c		
5	Cell, End	2		15-1/2	9	15/16	PPP-F-320	CF	WR	SW	V3c		
6	Cell, Top and Bottom	2		19-7/8	15-1/2	1-1/2	PPP-F-320	CF	WR	SW	V3c		
7	Box,Fiberboard	1		19-7/8	15-1/2	12	PPP-B-636	CF	WR	SW	V3c		

Lev. A - PACKED WEIGHT
Lev. A - PACKED VOLUME

T - Type
C - Class
V - Variety
G - Grade
F - Flute
S - Style

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APPENDIX A

RELIABILITY TEST PLAN FOR RADIO SET AN/GRC-106

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10.0 SCOPE - This test plan defines the general requirements, responsibilities, details, and operations necessary to conduct the Reliability Test Program for the AN/GRC-106( ) Radio Set.

20.0 PURPOSE - The purpose of this test is to provide the information necessary to conduct the Reliability Test Program for the AN/GRC-106( ) Radio Set by subjecting the equipment to an extended period of monitored operation under the conditions specified herein. The test program includes accept/reject criteria for production lots.

30.0 APPLICABLE DOCUMENTS - The following publication forms a part of this document to the extent referenced herein.

MIL-STD-781 Reliability Tests, Exponential Distribution.

40.0 RELIABILITY REQUIREMENTS - The contractor shall conduct a reliability acceptance test on each month's production based on a specified mean-time-between-failure (MTBF) of one hundred twenty five (125) hours. The test shall be conducted in accordance with Test Plan III of MIL-STD-781. Equipment quantities selected for reliability testing shall be in accordance with paragraph 4.2.4 of MIL-STD-781: except that the quantity per monthly lot shall be three (3) units.

40.1 TEST PREREQUISITES -

40.1.1 Acceptance of Equipment for testing - All equipment selected for reliability testing shall be from lots that have been subjected to and met Group A and B inspection. Performance data from such tests on units selected for reliability testing shall be made part of the reliability report.

40.1.2 Test Time Indicators - Each radio set shall be provided with an elapsed time indicator which shall accurately show accumulated test time and point of equipment failures or malfunctions.

40.1.3 Burn-in/Debugging - Each radio set shall have been subjected to the specified preconditioning paragraphs 3.5 and 4.6 to exclude the early failure period. Failures that occurred during this period will be considered non-relevant, but shall be recorded and reported for informational purposes.

40.1.4 Test Facility - The test facility shall be in accordance with paragraph 5.3 of MIL-STD-781 to the extent required by this test plan. The test equipment and support facility shall be in proper operating condition and calibration prior to the commencement of testing. All test equipment must be maintained in proper calibration throughout the test.

40.1.5 Test Equipment Required - A list of test equipment shall be supplied indicating manufacturers' type, serial number and due calibration date. This list shall be made part of the production test plan.

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40.2 TEST PROCEDURES - The following test shall be performed on all units each twenty-four (24) hours.

Electrical Tests	Req Para	Insp Para
Receiver		
Over-all Gain	3.6.11	4.8.12
Noise Figure	3.6.12	4.8.13
Audio Distortion	3.6.31	4.8.33
Audio Power Output	3.6.32	4.8.34
Metering	3.6.41	4.8.43
Transmitter		
Intermodulation Distortion	3.6.3	4.8.4
Hum and Noise	3.6.4	4.8.5
Carrier Suppression	3.6.6	4.8.7
Opposite Sideband Suppression	3.6.7	4.8.8
Short Term Stability	3.6.21.2	4.8.23.2
Frequency Accuracy	3.6.22	4.8.24
SSB Power Output	3.6.33.4	4.8.35.4
CW and FSK Power Output	3.6.33.6	4.8.35.6
Metering	3.6.41	4.8.43

Noise figure and power output shall be monitored during the duty cycle and during the last ten (10) minutes of every hour during vibration.

40.3 Environmental and Operational Stresses - The vibration and input voltage cycling requirements shall be employed as stated in Figure A7 of this specification.

40.3.1 Temperature Test Condition - 40°C ± 5°C

40.3.2 Vibration Test Condition - 2.2G ± 10% peak acceleration value at any non-resonant frequency between 20 and 55 Hz measured at the mounting points on the equipment. The duration of vibration shall be at least ten (10) minutes during each hour of equipment operating time.

40.3.3 Input Voltage Cycling Test Condition - The units shall be operated at 30.0 plus 0.0 minus 0.5 volts dc for the first 20% of the "on" time. The next 60% of the "on" time shall be 27.0 ± 0.5 volts dc and the remaining 20% shall be at 24.0 plus 0.5 minus 0.0 volts dc as shown in figure A7.

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40.3.4 ON-OFF Cycle Test Condition - The "ON-OFF" cycle shall be in accordance with figure A7. The equipment ON-OFF cycle, Duty Cycle, Vibration and Frequency sequencing and their relationship are shown in Figure A7. This sequence shall be followed throughout the test.

40.4 Test Condition -

40.4.1 Cycle - The testing shall be conducted on a twenty-four (24) hour, seven (7) days-per-week basis. Paragraph 5.4.7 of MIL-STD-781 shall apply.

40.4.2 Duty Cycle - The Transmit-receiver cycling shall be in accordance with Figure A7.

40.4.3 Test Frequencies - The radio sets shall be cycled sequentially through the prescribed frequencies of 4.6.2 in accordance with procedure of Figure A7. The operating frequencies of each radio set shall be such that the operation of one radio set will not interfere with testing of another.

40.4.4 Test Measurements - Measurements during reliability testing and criteria of satisfactory performance shall be as described in paragraph 50.2. The full sequence of tests specified in paragraph 50.5 herein shall be performed a minimum of once for each twenty-four (24) hour period on each radio set as soon as practical after thermal equilibrium has been reached during the "ON" period of the on-off cycling. These test measurements shall be taken at a successively higher frequency each day in accordance with the schedule of paragraph 4.6.2. The ten (10) minutes vibration shall not be suspended during the time test measurements are taken. The sequence of selecting equipment for measurements shall be such as to assure that measurements for each parameter on each unit are taken at least once under vibration. All data taken while the equipment is vibrating shall be so indicated with an asterisk (*). Data taken during the high and low voltage portion of the ON-OFF cycle shall be so indicated with an "H" or "L" respectively.

40.4.5 Monitoring - The radio sets shall be monitored to detect catastrophic failure or major degradation in transmit or receive operation. The operating parameters of Power Output and Noise Figure shall be continuously monitored during all periods of ON time so that any catastrophic malfunction of the equipment will be detected at the time of occurrence.

40.4.6 Preventive Maintenance - No preventive maintenance shall be performed on the units during reliability testing except as may be scheduled and approved by the responsible Government technical activity.

40.5 Failure Reporting and Analysis - A failure reporting and analysis system shall be implemented.

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40.6 Accept-Reject Criteria - The accept-reject criteria shall be per Test Plan III of Standard MIL-STD-781.

50.0 PERFORMANCE

50.1 Test Prerequisite Procedures

50.1.1 Selection of Equipment - Selection of equipment shall be arbitrary from the monthly production lot. Paragraph 5.4.1 of MIL-STD-781 shall otherwise apply.

50.1.2 Installation of Equipment - The equipment to be tested and necessary instrumentation to meet the requirements for the test shall be installed to provide for the safety of the equipment, the test facility and the personnel.

50.1.3 Test-Time Indicator - The elapsed time indicator shall be located in the system test fixture and shall be activated by the 27 volt output from the radio set. The 27 volt line is energized only when the radio set is in one of the operational modes and is de-energized in the OFF, STAND-BY and OVEN positions. Therefore, test time cannot be accumulated on the radio set unless it is operating.

50.1.4 Pre-Test Measurements - The pre-test measurements shall be performed in accordance with the electrical test procedure of paragraph 50.5 herein. Measurements shall be taken at each frequency of paragraph 4.6.2 while the radio set is being cycled through the OFF-ON, vibration and input voltage conditions as specified in paragraph 40.3.

50.2 Minimum Performance Limits - The minimum performance limits shall be in accordance with the applicable paragraphs of section 3.6 of this specification.

50.3 Test Frequencies - The test frequencies used during the reliability test shall be those listed in paragraph 4.6.2.

50.4 Test Equipment - The test and monitoring equipment for the implementation of this reliability program is as follows: Tolerances shall be as specified by the instrument manufacturer and substitution of equivalent instrumentation is permitted.

Distortion Analyzer and Audio Meter - Hewlett-Packard 330D
Signal Generator - Rhode & Schwartz HS1205/1206
Noise Source - Rhode & Schwartz SKTU Code BN-4151/2/75
RF VTVM - Boonton 91C with 50-OHMS Load and HI-Z Probe
System Test Fixture, AN/GRC-106 - General Dynamics
50-OHMS - 400 watt load

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VTVM with T-Probe - Hewlett-Packard 410B or equivalent
Standard Test Message Source
Two-tone Audio Generator - Panoramic TTG-2
Oscilloscope - Tektronix 581 with 82 Plug-in
Panalyzor - Singer Metrics - SSB-3B
Frequency measuring equipment and master frequency standard
(Stability one part in 10^9 per day or better)
Frequency Counter - Hewlett-Packard 524D
0-120 db Attenuators - Hewlett-Packard 355C and 355D

50.5 Test Procedures - The following performance parameters will be measured for each receiver-transmitter, once each twenty-four (24) hours. The performance limits, below which a failure shall be deemed to have occurred, shall be as referenced in paragraph 50.2 above. The test time for each radio set shall be measured by the elapsed time meters which record the equipment's "ON" time. The equipment shall be operated on a seven (7) day per week, twenty-four (24) per day basis until an accept/reject decision has been reached.

NOTE-TUNE-UP Procedure - Each Time the Frequency Control Knobs are turned, the following sequence must be followed if proper operation is to be obtained:

(A1) Set Antenna Tune and Load Controls to starting position as shown on chart for a 50-OHM doublet antenna.

(A2) Turn the TUNE/OPERATE switch on the AM-3349 to the Tune position. This provides a reduced signal for Tuning and Loading.

(A3) Alternately adjust the Tune and Load so that both the Tune and Load meters are centered. More than one adjustment may be required since on some frequencies there is interaction between these two controls. When the tuning is complete, return the TUNE/OPERATE switch to OPERATE.

50.5.1 Over-all Gain

50.5.1.1 Connect the equipment as shown in Figure A1 with the 2-watt output of the Test Fixture connected to the meter input of the Distortion Analyzer.

50.5.1.2 Set the controls of the RT-834 as follows:

Service Selector	SSB/NSK
RF Gain	Full CW (maximum)
Audio Gain	Full CW (maximum)
Squelch	OFF
Frequency Vernier	OFF
Frequency Controls	Set to Test Frequency

50.5.1.3 Tune Signal Generator to a frequency of 1,000 Hz above the test frequency and adjust the output power to 4 microvolts.

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50.5.1.4 Record the voltmeter reading on the Data Sheet. Output must be at least 2.0 watts (34.6 volts, 600 OHMS).

50.5.1.5 Repeat the above procedure and obtain readings for 19 additional frequencies of paragraph 4.6.2 including the asterisked frequencies. Record on Data Sheet.

50.5.2 Audio Power Output

50.5.2.1 With the equipment connected and adjusted in accordance with paragraph 50.5.1.1 through 50.5.1.3 above, return the RT-834 frequency control to the test frequency. Enter this frequency on the Data Sheet.

50.5.2.2 Turn the Audio Gain Control to verify that a minimum of 34.6 volts (2 watts) of audio output power can be obtained in the CW direction and a maximum of .77 volts (nominal 0) can be obtained in the CCW direction. Record on Data Sheet.

50.5.3 Audio Distortion and Metering Check

50.5.3.1 Connect and adjust the equipment in accordance with paragraphs 50.5.1.1 through 50.5.1.3 above, except that the Audio Gain Control shall be set full CCW (minimum).

50.5.3.2 Increase the Signal Generator output voltage to 1.0 volt and note the indication of increased received power on the Receive Meter. Check on Data Sheet.

50.5.3.3 Increase the Audio Gain Control to obtain an output of 34.6 volts (2 watts).

50.5.3.4 Change the connection on the Distortion Analyzer to the Distortion Input and measure the total harmonic distortion of the audio output as a percent of the reference level. Record on Data Sheet.

50.5.3.5 Change the connection on the Test Fixture from the 2-watt output to the 10-MW output and change the connection on the Distortion Analyzer from Distortion Input to Meter Input.

50.5.3.6 Adjust the Audio Gain Control of the RT-834 to obtain an output of 2.45 volts (10 MW/600-OHMS). Repeat the measurement of paragraph 50.5.3.4 and record on the Data Sheet.

50.5.4 Noise Figure

50.5.4.1 Connect the equipment as shown in Figure A2.

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50.5.4.2 Set the Controls of the RT-834 as follows:

Service Selector	SSB/NSK
RF Gain	Full CW (maximum)
Squelch	OFF
Frequency Vernier	OFF
Frequency Control	Set to Test Frequency

50.5.4.3 Set the controls of the Noise Generator as follows:

Range Switch	15-DB
Output Control	Full CCW

50.5.4.4 Note the VTVM reading (RF output noise).

50.5.4.5 Increase the Noise Generator output until the VTVM reading increases 3-DB. Read the Noise Generator Meter directly in DB and record on the Data Sheet. Noise figure shall be not more than 12 DB.

50.5.5 CW Power Output and Metering Check

50.5.5.1 Connect the test equipment as shown in Figure A3.

50.5.5.2 Set the Frequency Controls to the Test Frequency and the Service Selector switch to the CW position. Tune the radio set in accordance to "Note" of paragraph 50.5.

50.5.5.3 Key the radio set and record the RF Voltmeter reading on the Data Sheet. The reading shall be 100 volts plus 12, minus 11 volts.

50.5.5.4 Switch the test meter to the following positions and note the readings:

High Voltage
Low Voltage
Prime Voltage
Driver Current

The test meter should indicate within the dark green (triangle) area. Switch the test meter to the following positions and note the readings:

Grid Drive
PA Current
Power Output

The test meter should indicate within the light grey area. Check on the Data Sheet.

50.5.6 SSB Power Output

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50.5.6.1 Connect the equipment as shown in Figure A4.

50.5.6.2 Set the Frequency Control to the Test Frequency, the Service Selector switch to SSB/NSK, and the VOX switch to Push-to-talk.

50.5.6.3 Select the two-tone generator input on the system test fixture and, using the meter in the Distortion Analyzer, set the two-tone Generator for the following outputs:

Tone A: 1.5-KC, 200-MV
Tone B: 2.5-KC, 200-MV

50.5.6.4 Key the transmitter, read the output power on the RF VTVM, and record on the Data Sheet. The reading shall be 141 volts plus 37, minus 15-volts.

50.5.6.5 With the two-tone generator adjust in accordance with paragraph 50.5.6.3, observe the output of the two-tone generator on the oscilloscope and establish a 4 centimeter reference. Switch the system test fixture and the oscilloscope to the standard test message and adjust the peak amplitude of the message to the 4 centimeter reference. Switch the system test fixture to the two-tone input and observe the transmitter output waveform on the oscilloscope and establish a 4 centimeter reference. Switch the input of the system test fixture to the standard test message. Observe that the peak amplitude of the transmitted standard test message is at least 4 centimeters. Check on the Data Sheet.

50.5.7 Carrier Suppression, Opposite Sideband Suppression, Hum and Noise and Intermodulation Distortion.

50.5.7.1 With the equipment connected and adjusted in accordance with paragraphs 50.5.6.1, 50.5.6.2, and 50.5.6.3, key the transmitter and establish the two-tone log reference on the panalyzor using the 3.5-KC or 5-KC sweep width.

50.5.7.2 Observe that the carrier leakage is suppressed a minimum of 50-DB below wither of the two test tones. Check on the Data Sheet.

50.5.7.3 Increase the panalyzor sweep width to 14-KC and observe that the opposite sideband is suppressed a minimum of 50-DB below either of the two test tones. (NOTE: This is opposite sideband only and intermodulation products are excepted). Check on the Data Sheet.

50.5.7.4 Observe that IM Products are at least-35 DB down from either of the test tones. Record actual value on Data Sheet.

50.5.7.5 Reduce the panalyzor sweep width to 3.5-KC or lower and observe that the hum and noise levels are a minimum of -50-DB below the output of either of the two test tones. Check on the Data Sheet.

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50.5.8 Frequency Stability

50.5.8.1 Connect the equipment as shown in Figure A5.

50.5.8.2 Using the frequency measuring equipment, determine the error at 1-MC and record on the Data Sheet. Determine that this reading meets the stability requirements of five parts in 10^7 per hour after 1 hour of warmup.

NOTE: To determine the error at 1-MC, divide the actual error at 5-MC by 5.

50.5.9 Frequency Accuracy

50.5.9.1 Connect the equipment as shown in Figure A6.

50.5.9.2 Set the service selector switch to AM (with no modulation) and the frequency controls to the test frequency. Using a high impedance resistive network connected across the 50-OHM load, feed approximately 3 volts of RF into the frequency counter.

50.5.9.3 Record the counter reading on the Data Sheet, Column 62. Verify that the counter reading (carrier frequency) does not differ from the dial setting by more than (the difference between the internal standard and 5-MC) times (the dial setting divided by 5-MC). Check on the Data Sheet.

50.6 Environmental and Operational Stresses - The environmental and operational stress facilities will be provided as follows:

50.6.1 Temperature Test Condition - A thermostatically controlled ambient temperature of 40°C ± 5°C shall be provided for the test area.

50.6.2 Vibration Test Condition - Vibration tables, located at the test area, will provide the vibration of ±2.2G at any non-resonant frequency between 20 and 55 Hz measured at the mounting points on the equipment. The vibration tables will be energized for ten (10) minutes out of each hour of "ON" time.

50.6.3 Input Voltage Cycling Test Condition - A fused power supply will provide the input voltages of 24-30 volts dc.

50.6.4 ON-OFF Cycle Test Condition - The units shall be automatically programmed for the six (6) hour ON-OFF cycle described in paragraph 40.3.4.

50.7 Test Conditions

50.7.1 Cycle - The testing will be conducted on a twenty-four (24) hour seven (7) days-per-week basis to determine accept-reject criteria in conformance with paragraph 40.7.

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50.7.2 Monitoring - The audio and radio frequency outputs of the radio sets will be automatically monitored for catastrophic failure or major degradation in transmit or receive operation. When a failure in either output occurs, an alarm system will be activated which will turn off the input voltages and the elapsed operating time meter. Failure times for other parameters shall be in accordance with paragraph 5.4.6 of MIL-STD-781.

50.7.3 Maintenance Measures to be performed - No adjustment or repair will be made on the radio set except that necessary to enable a failed radio set to perform as specified. Operator controls shall be adjusted as required.

50.8 Test Records

50.8.1 Operating Time Log - An operating time log shall be maintained for each unit starting with the initial receipt of the radio set and continuing through the completion of testing. The following data shall be recorded:

Elapsed time meter reading
Operate time
Accumulated time
Name of test or operation
Failure report number (if applicable)
Signature of operator

These entries will provide the necessary elapsed time information, the chronology of the test sequences and a record of all test interruptions. In addition, the operator shall enter the reason for any test interruption and any significant events or notes applicable to the period of operation.

50.8.2 Failure Tag - When a radio set is removed from the test area for repair, a filure tag will be attached to the unit showing (in part) the failure report number, associated reports and the operation sheet page and line number.

50.8.3 Equipment Repair Sheet - A failed radio set will be sent to troubleshoot and repair where the radio set will be analyzed to isolate and repair the failure. The finding will be documented on the equipment repair sheet and the failure report.

50.9 Restoration of Failed Equipments - Paragraph 5.9 of Standard MIL-STD-781 shall apply.

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TEST SET-UP DIAGRAMS

(See Test Procedures Paragraph 50.5)

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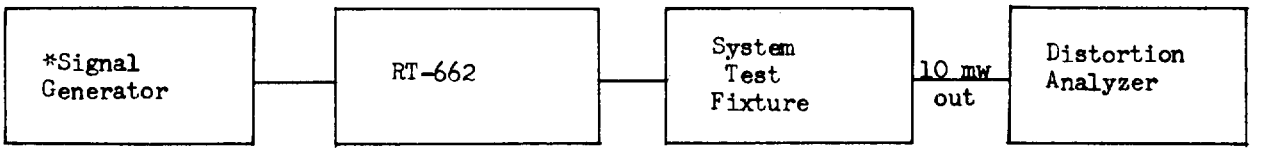


FIGURE A1

*This includes the necessary attenuators to adjust for the various signal levels and source impedance (75-ohms) required to perform this test.

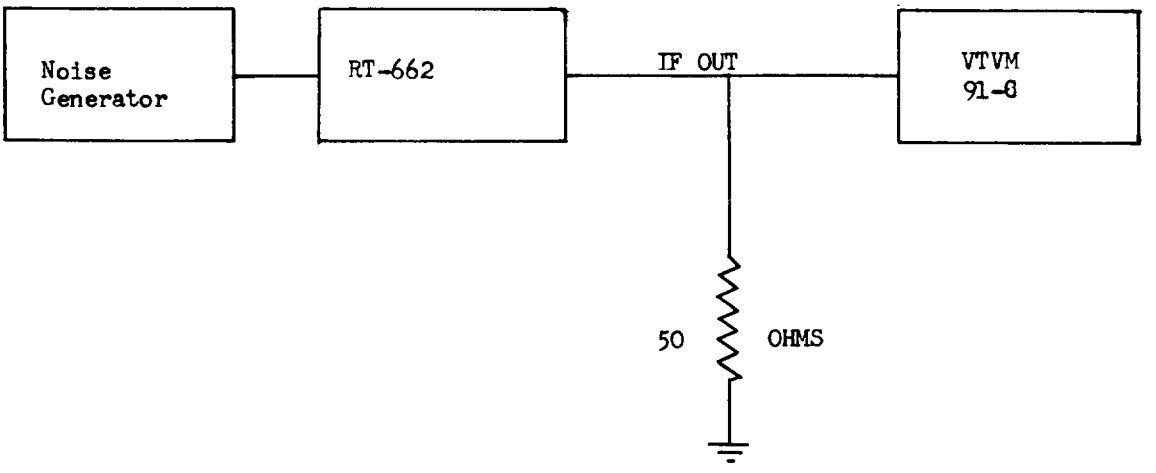


FIGURE A2

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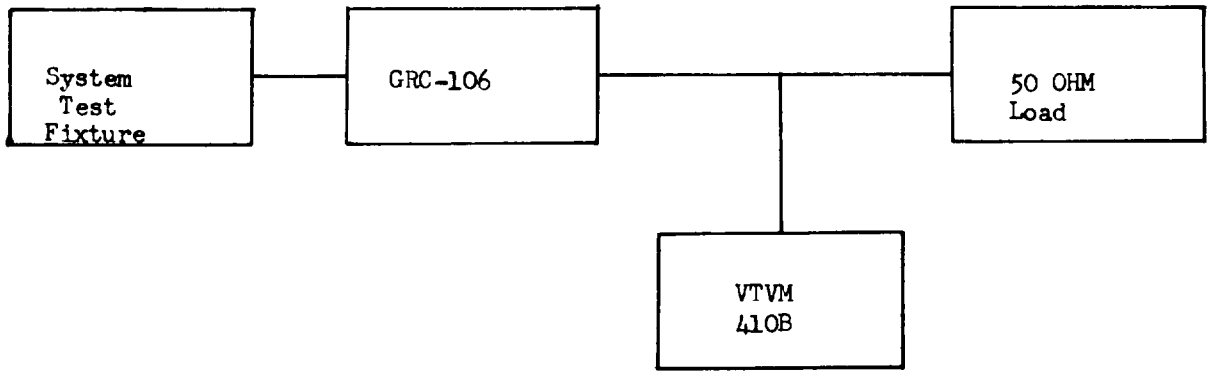


FIGURE A3

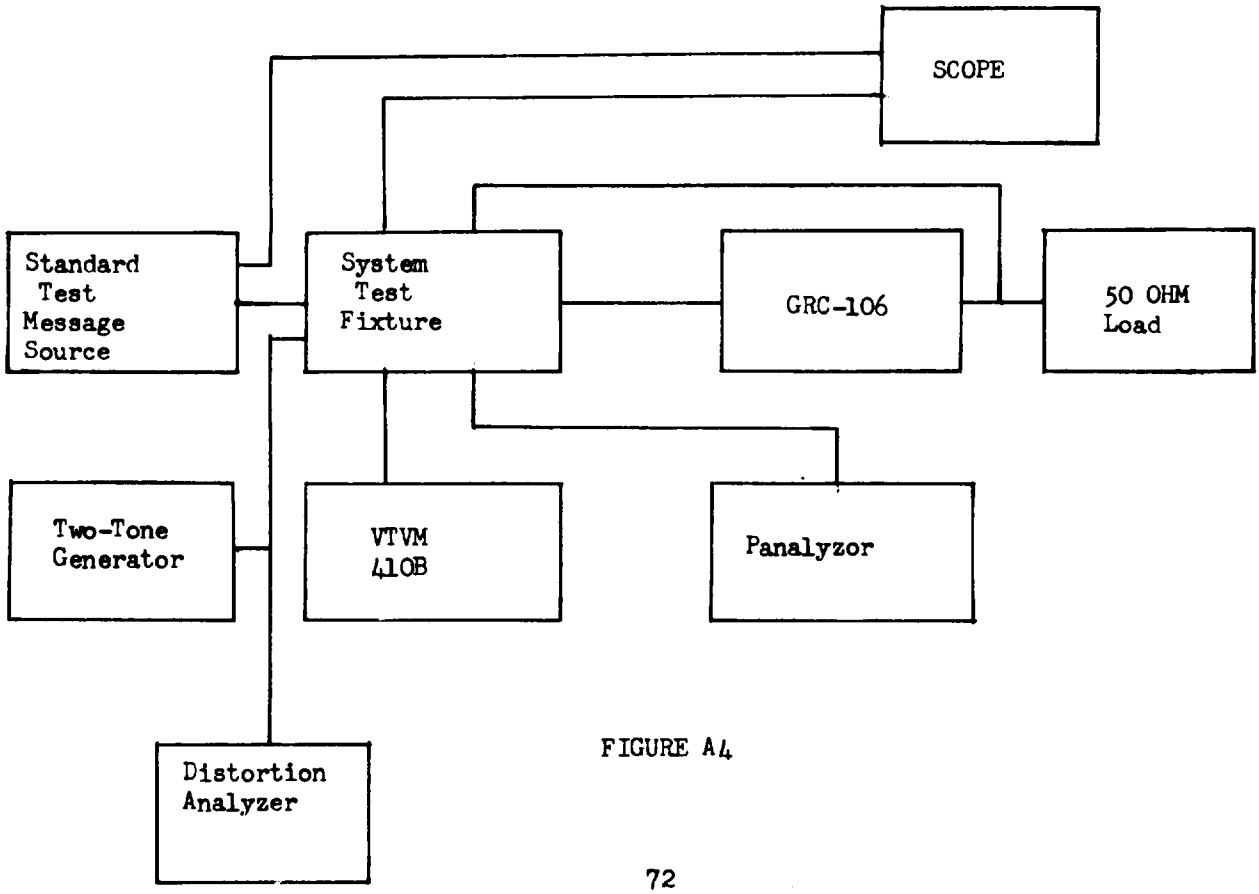


FIGURE A4

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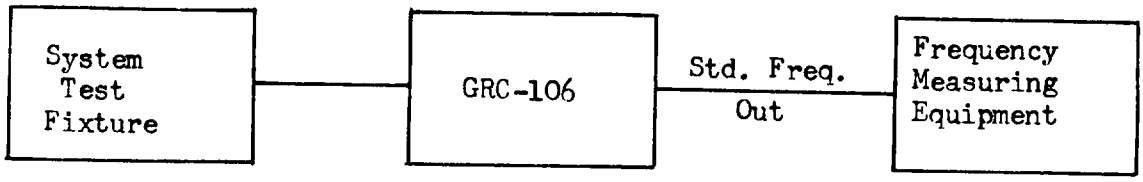


FIGURE A5

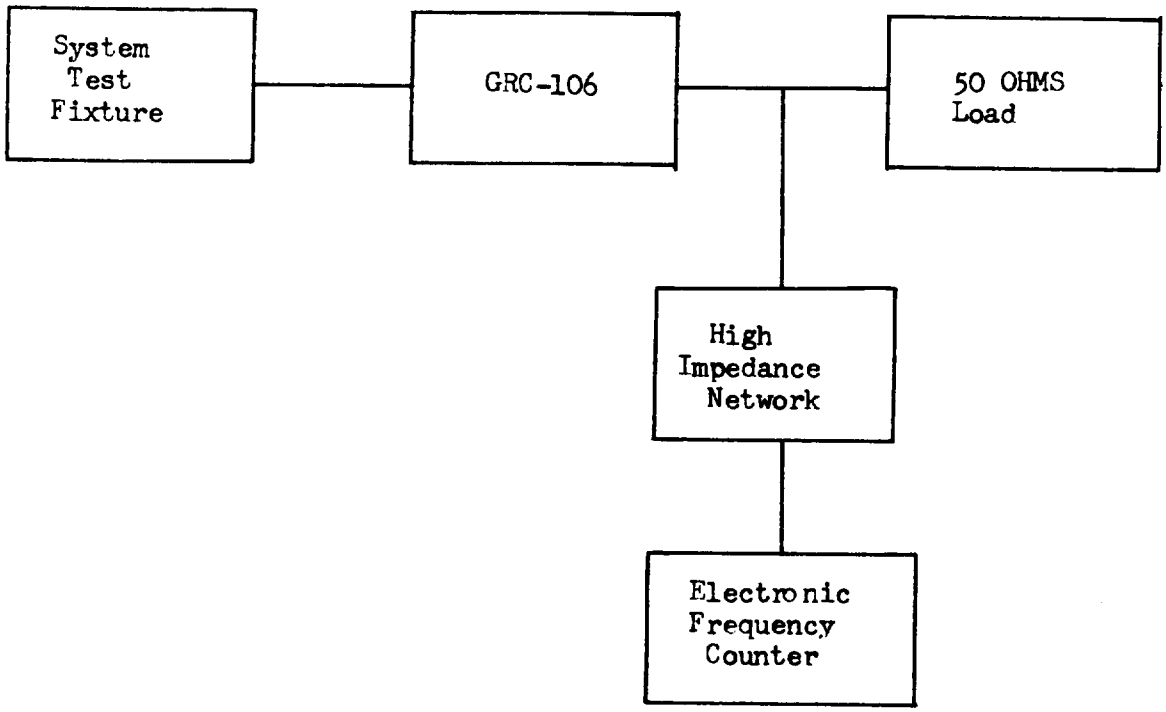


FIGURE A6

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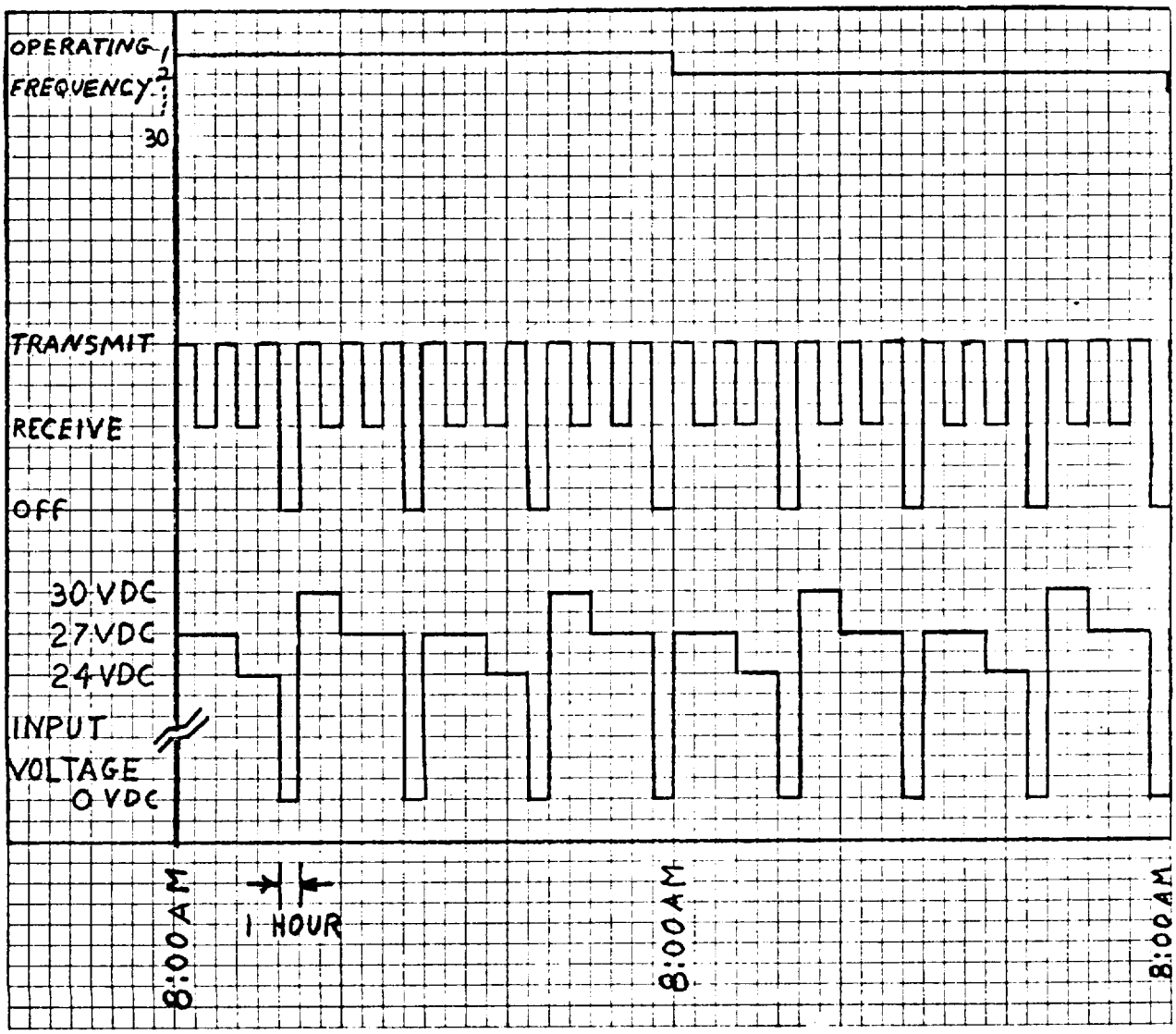


FIGURE A7
EQUIPMENT CYCLING

1. Vibrate equipment for 10 minutes during each transmit and receive period.

2. Once each day, special tests will be performed as follows:

a. Perform the tests 3 through 13 of Table A1. Each day, a different one of these tests shall be performed during the ten minute vibration period, repeating the sequence every eleven days.

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FIGURE A7 - EQUIPMENT CYCLING - (contd) -

b. After making the tests of a above, the two tone SSB power output and noise figure shall be measured at the 30 frequencies of paragraph 4.6.2.

c. The tests of a and b above shall be conducted at a different input voltage each day at the three levels shown for the input voltage cycling portion of figure A7 and this sequency repeated every three days.

3. Normal power output and noise figure tests made during the "Transmit-Receive-Off" cycle are suspended for the duration of the special tests.

4. During the normal transmit cycle, SSB Voice Power output shall be measured during the first ten (10) minutes and SSB two tone power output shall be measured during the remaining fifty (50) minutes. During the normal receive cycle, noise figure shall be measured.

5. Operating frequency cycling shall be performed using the frequencies of paragraph 4.6.2. At the end of each 24 hour period, the test frequency shall be increased to the next higher frequency of 4.6.2. When the test period for the highest frequency of 4.6.2 is completed, the frequency shall be changed to the lowest frequency of paragraph 4.6.2 and the cycling continued.

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Test Number	Test	Req Para	Test Para	Remarks
1	Workmanship	3.9		Performed at start and end of Reliability test only
2	SSB Voice & Two Tone	3.6.33.4	4.8.35.4	Measured at frequencies of Para 4.6.2
3	CW Power Output	3.6.33.6	4.8.35.6	Measured at frequencies of Para 4.6.2
4	Carrier Suppression	3.6.6	4.8.7	
5	Opposite sideband Suppression	3.6.7	4.8.8	
6	Hum & Noise	3.6.4	4.8.5	
7	IM Distortion	3.6.3	4.8.4	
8	Frequency Accuracy	3.6.22	4.8.24	Measure after at least one hour
9	Long term stability	3.6.21.1	4.8.23.1	" " "
10	Audio Power Output	3.6.32	4.8.34	
11	Audio Distortion	3.6.31	4.8.33	
12	Overall Gain	3.6.11	4.8.12	
13	Metering	3.6.41	4.8.43	
14	Noise Figure	3.6.12	4.8.13	Measure at frequencies of Para 4.6.2

TABLE A1

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OMB Approval
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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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